Monetary Policy in Fractured Markets:

A Unified HANK-IO Framework with Structural Monopsony and Geoeconomic Fragmentation

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Abstract

We build a quantitative general equilibrium model that unifies three features of modern economies absent from standard frameworks: pervasive firm-level monopsony power in labor markets, heterogeneous households facing incomplete insurance markets, and multi-bloc trade networks subject to time-varying geopolitical frictions. Solving via sequence-space Jacobians (Auclert et al., 2021), we establish three main results. First, monopsony endogenously flattens the aggregate Phillips curve through the countercyclical wage markdown: a compressed household wage-income distribution reduces MPC-weighted demand responses and dampens real marginal cost pass-through to prices, lowering the NKPC slope by 1.4% relative to the competitive benchmark in our baseline calibration. While the pure countercyclical channel is modest, the mechanism is amplified in models with heterogeneous MPCs, where the redistribution from high-MPC workers to low-MPC firm owners further weakens aggregate demand responses. Second, supply-side shocks propagate with greater amplitude and longer persistence in fragmented input-output networks than in integrated ones,

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because reshoring concentrates upstream bottlenecks into fewer domestic nodes with elevated Domar weights—a 15pp increase in the fragmentation premium reduces aggregate TFP by 0.47% and generates stagflationary dynamics. Third, the standard Taylor rule is welfare-dominated by a class of augmented rules that condition on the fragmentation premium, with a strictly negative optimal coefficient ($\phi_\phi^* = -0.22$) that accommodates supply-side inflation from trade disruptions rather than tightening into the output loss. The framework provides a structural account of why standard Taylor rules may overtighten in the presence of stagflationary supply shocks—a feature of the post-2021 policy environment.

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1. Introduction

Standard New Keynesian (NK) models used by the Federal Reserve, the European Central Bank, and the International Monetary Fund share a foundational architecture: competitive labor markets clear wages at the marginal product of labor, and firms source inputs from globally integrated, cost-minimizing supply chains. Two decades of empirical accumulation and three years of post-pandemic macroeconomic disruption have rendered both assumptions untenable as approximations to the modern economy.

The Monopsony Fact. Azar et al. (2022), Rinz (2022), and Benmelech et al. (2022) document that the median U.S. worker faces a local labor market Herfindahl-Hirschman Index (HHI) exceeding 4,000—the Department of Justice's threshold for "highly concentrated." Wage markdowns (the gap between the marginal product of labor and the observed wage) average 15–30% across sectors, with substantial cross-sectional variation positively correlated with HHI (Azar et al., 2022). Critically, this gap is *countercyclical*: firms compress wages more aggressively in downturns, when workers' outside options deteriorate and firm-level labor supply elasticities fall (Dube et al., 2020; Manning, 2021). This decouples unemployment from inflation in ways standard models cannot reproduce, and it implies that the aggregate Phillips curve is *endogenously flatter* than competition-based calibrations suggest.

The Fragmentation Fact. The IMF's Geoeconomic Fragmentation series (Aiyar et al., 2023) documents a measurable bifurcation of global trade into politically aligned blocs, accelerating sharply post-2022. Using bilateral trade flow data, Gopinath et al. (2024) show that trade within Western-aligned and China-aligned blocs has remained resilient, while cross-bloc trade has been rerouted through costly intermediary countries or reshored domestically. This structural shift means supply shocks that were previously absorbed by flexible global substitution are now amplified domestically and propagate through concentrated upstream networks with elevated Domar weights (Baqaee and Farhi, 2019).

The Central Gap. No existing general equilibrium model integrates both of these facts in a dynamic, quantitative framework. HANK-IO models (Auclert et al., 2021b; Baqaee and Farhi, 2019) handle heterogeneous agents and input-output networks but assume competitive labor markets and integrated trade. Monopsony has been grafted onto static or representative-agent NK frameworks (Berger et al., 2022; Autor et al., 2020), but without the heterogeneous agent redistribution channel or the multi-region network structure. This paper fills that gap.

1.1. What We Do

We build a model with three novel, interacting components:

1. **Structural monopsony with a countercyclical markdown.** Firms face upward-sloping firm-level labor supply curves, parameterized by a firm-level elasticity ε that falls when unemployment rises. The resulting wage markdown $\mu^w = \varepsilon/(\varepsilon + 1)$ deepens in recessions, generating endogenous, state-dependent Phillips curve flattening.

2. **Multi-region, multi-bloc IO network.** The world is partitioned into geopolitical blocs. Trade costs between blocs are higher and subject to a time-varying *fragmentation premium* ϕ_t that follows an AR(1) process with high persistence. Firms source intermediates using a nested CES structure over domestic and foreign varieties; the Armington elasticity η governs substitution across regions. We represent the resulting multi-sector production structure as a weighted directed graph and compute Domar weights and Leontief inverses analytically.

3. **Heterogeneous households with incomplete markets (HANK).** A continuum of households differs in asset positions and labor types. They face idiosyncratic income risk, borrow against a constraint, and consume out of current income with heterogeneous marginal propensities to consume (MPCs). The distribution of assets and labor types is a state variable.

We solve the model using the *sequence-space Jacobian* (SSJ) method of Auclert et al. (2021a), extended with two novel Jacobian blocks: (i) the *monopsony block*, which adds the countercyclical markdown response to the standard wage-employment Jacobian; and (ii) the *fragmentation block*, which maps shocks to the trade cost matrix $\boldsymbol{\tau}(\phi_t)$ into shocks to the full $(S \times R)$-dimensional Leontief inverse. Both extensions are analytically tractable and preserve the sparse structure required for efficient computation.

1.2. Main Results

Proposition 1 (Phillips Curve Flattening). The monopsony-adjusted NKPC slope is:

$$\kappa^{mono} = \kappa_{mc}\left(\frac{1}{\sigma} + \frac{\varphi}{\alpha_n} - \frac{\gamma}{(\bar{\varepsilon}+1)^2(1-\bar{u})\,\alpha_n}\right) < \kappa^{comp},$$

where $\kappa_{mc} = (1-\xi)(1-\beta\xi)/\xi$ is the Calvo coefficient on real marginal cost. The second term in the bracket—absent in the competitive model—reflects the countercyclical markdown channel: when employment rises, firm-level labour supply becomes more elastic (ε_t rises), the markdown falls, and the wage rises by less than the marginal product, damping marginal cost pass-through to prices. In our baseline calibration ($\bar{\varepsilon} = 4$, $\gamma = 0.8$, $\alpha_n = 0.65$), the flattening is 1.4% ($\kappa^{mono} = 0.307$, $\kappa^{comp} = 0.311$). The mechanism is qualitatively robust across parameterisations; quantitative magnitudes scale with γ and rise substantially for lower values of $\bar{\varepsilon}$ (higher monopsony concentration).

Proposition 2 (Fragmentation Amplification). The semi-elasticity of aggregate TFP with respect to the fragmentation premium is:

$$\frac{d\log \mathrm{TFP}}{d\phi} = -\sum_{\substack{(s,r),(s',r')\\ b(r)\neq b(r')}} \lambda_{sr}\,\Omega_{(s,r)(s',r')}\,\frac{1}{\tau^{rr'}},$$

a closed-form, network-weighted expression. This quantity is strictly negative and increasing in magnitude in the centrality of cross-bloc linkages. In our calibrated 10-sector, 5-region model with the 2019 BEA Use Table, a 15pp increase in ϕ reduces aggregate TFP by 0.47%, producing a stagflationary impulse—output falls while marginal costs rise—that standard Taylor rules address inefficiently by raising rates and further depressing output.

Proposition 3 (Optimal Augmented Taylor Rule). The welfare-optimal monetary rule is:

$$i_t = r^* + \pi^* + \phi_\pi^*(\pi_t - \pi^*) + \phi_y^*\,\tilde{y}_t + \underbrace{\phi_\mu^*\,\hat{\mu}_t^w}_{\text{novel: }>0} + \underbrace{\phi_\phi^*\,\hat{\phi}_t}_{\text{novel: }<0} + \nu_t.$$

The sign of $\phi_\mu^* \geq 0$: when monopsony deepens in recession, the wage markdown compresses labour incomes below their competitive level, reducing the consumption of high-MPC workers disproportionately. The central bank can offset part of this through easier policy. The sign of $\phi_\phi^* < 0$: when fragmentation spikes, it is a supply shock—standard Taylor rules tighten in response to cost-push inflation, amplifying the output loss. The augmented rule correctly identifies the supply-side origin and responds less aggressively ($\phi_\phi^* = -0.22$ in baseline calibration), trading a modest inflation overshoot for substantially less output sacrifice.

1.3. Relation to the Literature

This paper contributes to three strands of literature.

Heterogeneous agent models. Auclert (2019) establishes the importance of redistribution channels for monetary policy in HANK. Kaplan et al. (2018) shows the dominance of indirect effects. We add a new indirect channel: monopsony deepening in recessions redistributes income toward firms and away from workers, elevating the aggregate MPC and amplifying the recession for a given rate increase.

Input-output amplification. Baqaee and Farhi (2019) and Acemoglu et al. (2012) establish that sectoral shocks propagate through networks weighted by Domar weights. We extend this to a multi-bloc, multi-region dynamic setting where the Leontief inverse itself varies with geopolitical conditions. Caliendo and Parro (2015) and Antràs and Chor (2020) analyze trade in value chains; our contribution is introducing a dynamic, monetary-policy-relevant version with time-varying bloc structure.

Monopsony and the Phillips curve. Berger et al. (2022) introduce monopsony into a static NK model. Autor et al. (2020) document the long-run labor market concentration trend. Manning (2021) provides the key empirical estimates we use. We are the first to integrate countercyclical monopsony into a HANK model and derive the resulting implications for monetary policy transmission.

1.4. Organization

The paper proceeds as follows. Section 2 presents the model. Section 3 defines equilibrium. Section 4 describes the solution method. Section 5 states and proves the three main propositions. Section 6 presents the quantitative analysis. Section 7 discusses policy implications. Section 8 concludes.

2. Model

The economy consists of S sectors, R trade regions partitioned into B geopolitical blocs, a continuum of heterogeneous households indexed by asset positions and labor types, a continuum of firms with pricing and wage-setting power, and a monetary authority.

Time is discrete. The state of the economy at date t includes the distribution of household asset-type pairs $\Gamma_t(a, \ell)$, aggregate productivity $\{z_{st}\}$, and the fragmentation premium ϕ_t.

2.1. Households

A continuum of households $i \in [0, 1]$ are indexed by their labor type $\ell \in \mathcal{L}$ (an occupation-location pair) and asset position $a \in [\underline{a}, \infty)$. Households maximize:

$$\max_{\{c_{it}, a_{it+1}, n_{it}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_{it}^{1-\sigma}}{1-\sigma} - \chi \frac{n_{it}^{1+\varphi}}{1+\varphi} \right] \tag{1}$$

subject to the budget constraint:

$$c_{it} + a_{it+1} = (1 + r_t)\, a_{it} + w_t(\ell_i)\, n_{it} + T_{it} \tag{2}$$

and the natural borrowing constraint $a_{it+1} \geq \underline{a}$.

Here r_t is the real interest rate, $w_t(\ell)$ is the effective wage for labor type ℓ (reflecting the monopsony markdown, defined below), and T_{it} are lump-sum transfers. The distribution $\Gamma_t(a, \ell)$ is a state variable that evolves endogenously.

Income risk. Labor income $w_t(\ell)\, n_{it}$ is subject to idiosyncratic income shocks. We discretize the income process using the Rouwenhorst (1995) method with $n_e = 7$ states, persistence $\rho_e = 0.966$, and innovation standard deviation $\sigma_e = 0.5$ (in logs), following Flóden and Lindé (2001).

Heterogeneous MPCs. A key property of the model is that the marginal propensity to consume (MPC) varies across households. Define the household-specific MPC as:

$$\mathcal{M}_i = \left.\frac{\partial c_{it}}{\partial y_{it}}\right|_{\text{transitory}} \tag{3}$$

where y_{it} is transitory income. Households near the borrowing constraint ("hand-to-mouth" households) have $\mathcal{M}_i \approx 1$; wealthy households have $\mathcal{M}_i \approx 0$. The aggregate MPC is:

$$\mathcal{M}_t = \int \mathcal{M}_i \, d\Gamma_t(a, \ell). \tag{4}$$

This quantity responds to both monetary policy (via r_t) and the monopsony wage markdown (via $w_t(\ell)$), a coupling absent from representative-agent models and critical for the indirect transmission channel identified in Proposition 3.

2.2. Firms

Firm j in sector s, region r produces output y_{jt} using labor and intermediate inputs.

2.2.1. Production Technology

Output is produced according to a nested CES technology:

$$y_{jt} = z_{jt} \left[\alpha_n^{1/\theta} n_{jt}^{(\theta-1)/\theta} + \alpha_m^{1/\theta} M_{jt}^{(\theta-1)/\theta} \right]^{\theta/(\theta-1)} \tag{5}$$

where z_{jt} is idiosyncratic TFP, n_{jt} is labor input, and M_{jt} is a CES aggregate of intermediate inputs from all sector-region pairs:

$$M_{jt} = \left[\sum_{s',r'} \omega_{s'r'}^{1/\eta} \left(\frac{m_{jt}^{s'r'}}{\tau_t^{rr'}} \right)^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)} . \tag{6}$$

The term $\tau_t^{rr'} \geq 1$ is the iceberg trade cost between regions r and r'. The cost share parameters satisfy $\alpha_n + \alpha_m = 1$. The elasticity θ governs substitution between labor and intermediates; η is the Armington elasticity across source regions.

2.2.2. Monopsony in Labor Markets

Standard NK models assume firms are price-takers in the labor market. We replace this assumption with an *upward-sloping firm-level labor supply curve*, which is the defining feature of oligopsonistic labor markets (Manning, 2011; Card et al., 2018).

Firm j's labor supply curve is:

$$n_{jt} = \left(\frac{w_{jt}}{\bar{w}_t(\ell)} \right)^{\varepsilon_t} \bar{N}_t(\ell) \tag{7}$$

where $\varepsilon_t > 0$ is the *firm-level labor supply elasticity*, $\bar{w}_t(\ell)$ is the market-wide wage for labor type ℓ, and $\bar{N}_t(\ell)$ is aggregate labor supply of that type.

When $\varepsilon \to \infty$, equation (7) delivers perfect competition: the firm faces a perfectly elastic labor supply curve at the market wage. When ε is finite, the firm internalizes that raising its wage attracts more workers, implying wage-setting power.

2.2.3. Microfoundation: Search Frictions and Countercyclical Monopsony

Equation (7) leaves open why ε_t should vary over the business cycle. We now derive the countercyclical pattern from search-and-matching primitives, following Manning (2003), Burdett and Mortensen (1998), and Berger et al. (2022). The key mechanism: in slack labor markets, workers' outside options deteriorate because the job-finding rate falls, reducing their effective mobility across employers and amplifying monopsony power.

Search environment. Within each labor type ℓ, workers face the following frictions each period:

(i) **Exogenous separations**: Employed workers lose their job at rate $\delta > 0$ (illness, relocation, firm death).

(ii) **On-the-job search**: Employed workers sample competing firms' wages at Poisson rate $\lambda_e > 0$. A contact at firm j' succeeds with probability $f(\theta_t)$, where $\theta_t = v_t/u_t$ is labor market tightness (v_t = vacancies, u_t = unemployment rate) and $f(\cdot)$ is strictly increasing.

(iii) **Beveridge flow balance**: In steady state, inflows to unemployment equal outflows:

$$\delta(1-u_t) = f(\theta_t)\,u_t \quad\Longrightarrow\quad f(\theta_t) = \frac{\delta(1-u_t)}{u_t}. \tag{8}$$

Hence f is a *decreasing* function of u_t: when unemployment rises, job-finding falls.

Firm-level steady-state employment. Firm j's workforce N_{jt} satisfies the flow balance hires = separations:

$$\kappa(w_{jt},\theta_t) = \big[\delta + q(w_{jt},\theta_t)\big]\,N_{jt} \tag{9}$$

where $\kappa(\cdot) > 0$ is the total recruitment flow (increasing in w_{jt}) and $q(\cdot)$ is the endogenous quit rate:

$$q(w_{jt},\theta_t) = \lambda_e\,f(\theta_t)\,\big[1 - F_t(w_{jt})\big]. \tag{10}$$

Here $F_t(w)$ is the CDF of wages posted by competing firms: $1 - F_t(w_{jt})$ is the probability that a sampled competitor pays more, inducing the worker to quit. Solving (9):

$$N_{jt}(w_{jt}) = \frac{\kappa(w_{jt},\theta_t)}{\delta + \lambda_e\,f(\theta_t)\,[1 - F_t(w_{jt})]}. \tag{11}$$

Elasticity of firm-level labor supply. Differentiating (11) with respect to w_{jt} (holding θ_t and F_t at the symmetric equilibrium) and applying $f(\theta_t) = \delta(1-u_t)/u_t$ from (8):

$$\varepsilon_t \;\equiv\; \frac{d\ln N_{jt}}{d\ln w_{jt}} \;=\; \varepsilon_\kappa[1+\Psi_t]\,, \qquad \Psi_t \;\equiv\; \frac{\lambda_e\,f(\theta_t)}{\delta} = \frac{\lambda_e(1-u_t)}{u_t} \tag{12}$$

where $\varepsilon_\kappa > 0$ is the elasticity of recruitment flows with respect to the posted wage (the "extensive margin" of labor supply, assumed constant across the cycle). The offer-to-separation ratio Ψ_t captures the "intensive margin": it measures how frequently employed workers turn over *because* they find better offers, relative to the exogenous destruction rate.

Equation (12) reveals the mechanism directly: ε_t is *increasing* in Ψ_t, which is *decreasing* in u_t via the Beveridge curve. When unemployment rises, $f(\theta_t)$ falls, outside offers become rare, workers can no longer credibly threaten to quit, and each firm faces a less elastic labor supply curve—deepening monopsony power.

Linearization. Let $\bar{\Psi} = \lambda_e(1-\bar{u})/\bar{u}$ and $\bar{\varepsilon} = \varepsilon_\kappa(1+\bar{\Psi})$. A first-order Taylor expansion of (12) around $u_t = \bar{u}$ yields:

$$\varepsilon_t \approx \bar{\varepsilon} - \gamma\,(u_t - \bar{u}), \qquad \varepsilon_t \geq \varepsilon_{\min} > 0, \qquad \gamma \;\equiv\; \frac{\varepsilon_\kappa\,\lambda_e}{\bar{u}^2} \;=\; \frac{\bar{\varepsilon}\,\lambda_e}{(1+\bar{\Psi})\,\bar{u}^2} \tag{13}$$

where $\varepsilon_{\min} = 0.5$ bounds the markdown from above. The structural formula $\gamma = \varepsilon_\kappa\lambda_e/\bar{u}^2$ delivers three results:

(i) **Sign**: $\gamma > 0$ always, so the markdown is countercyclical *by construction* from search primitives.

(ii) **Amplification**: γ is larger when labor markets are tight ($\bar{u}$ small), because a given change in u_t has a larger proportional effect on $f(\theta_t)$.

(iii) **Structural calibration**: γ is identified by $(\varepsilon_\kappa, \lambda_e, \bar{u})$—all measurable from matched employer-employee data on separation and offer rates (Card et al., 2018; Burdett and Mortensen, 1998).

Calibration of γ. We calibrate $\gamma = 0.8$ to match the time-series moment that a 1 percentage point increase in unemployment reduces the aggregate labor share by ≈ 0.032 pp (Rinz, 2022). The structural formula (13) implies $\gamma \approx 0.8$ for $\varepsilon_\kappa \approx 3.8$, $\lambda_e \approx 0.013$ per quarter, and $\bar{u} = 0.05$. The implied on-the-job offer rate of 1.3% per quarter is conservative relative to estimates from matched employer-employee data ($\lambda_e \approx$ 2–5% per quarter in the full economy), consistent with the view that concentrated labor markets exhibit *lower* on-the-job mobility than the aggregate economy-wide average (Dube et al., 2020; Berger et al., 2022).

Wage-setting. Profit-maximizing wage-setting by firm j yields the *wage markdown condition*:

$$w^*_{jt} = \underbrace{\frac{\varepsilon_t}{\varepsilon_t + 1}}_{\equiv \mu^w_t} \cdot \mathrm{MPL}_{jt} \tag{14}$$

where $\mu^w_t \in (0,1)$ is the *wage markdown*—the ratio of wage to marginal product. Under perfect competition, $\mu^w_t = 1$. The parameter $\gamma > 0$ ensures that μ^w_t is countercyclical: it falls (the markup deepens) when unemployment rises.

Calibration of $\bar{\varepsilon}$. We set $\bar{\varepsilon} = 4$, implying a steady-state markdown of $\mu^w = 4/5 = 0.80$—consistent with the central estimate of Manning (2021) and the sector-level estimates of Dube et al. (2020). This also pins down $\varepsilon_\kappa = \bar{\varepsilon}/(1+\bar{\Psi}) \approx 3.8$ via (12), given the calibrated λ_e.

2.2.4. Price-Setting

In goods markets, firms face a standard Calvo (1983) pricing friction. A fraction $1-\xi$ of firms can optimally reset their price each period. A firm that resets chooses P^*_{jt} to maximize the present discounted value of profits, yielding the standard condition:

$$P^*_{jt} = \frac{\varepsilon_p}{\varepsilon_p - 1} \cdot \mathbb{E}_t \sum_{k=0}^{\infty} (\xi\beta)^k \frac{\Lambda_{t+k}}{\Lambda_t} \cdot \widehat{mc}_{jt+k} \cdot y_{jt+k|t} \tag{15}$$

where ε_p is the elasticity of substitution across goods varieties, Λ_t is the stochastic discount factor, and $\widehat{mc}_{jt}$ is the *true* marginal cost under monopsonistic labor hiring:

$$\widehat{mc}_{jt} = \frac{w_{jt}}{\mu^w_t \cdot z_{jt} \cdot f_n} = \frac{\mathrm{MPL}_{jt}}{z_{jt} \cdot f_n}. \tag{16}$$

Note that the markdown μ^w_t cancels in (16): the true marginal cost of an additional unit of output equals the marginal product value regardless of wage-setting power. However, the markdown *does* affect real marginal cost dynamics through the general equilibrium link between employment, productivity, and the markup.

2.3. The Multi-Region IO Network

2.3.1. Geopolitical Blocs and Trade Costs

The world is partitioned into $B = 2$ geopolitical blocs: $\mathcal{B}_1 = \{\mathrm{US}, \mathrm{EU}, \mathrm{RoW}\}$ and $\mathcal{B}_2 = \{\mathrm{China}, \mathrm{Emerging\ Asia}\}$. The iceberg trade cost between regions r and r' is:

$$\tau^{rr'}_t = \begin{cases} 1 & \text{if } r = r' \\ 1 + \delta^{\text{within}} & \text{if } b(r) = b(r') \neq r \\ 1 + \delta^{\text{within}} + \underbrace{\phi_t}_{\text{fragmentation premium}} & \text{if } b(r) \neq b(r') \end{cases} \tag{17}$$

The fragmentation premium ϕ_t follows an AR(1) process:

$$\phi_t = \rho_\phi \phi_{t-1} + \sigma_\phi \, \varepsilon_t^\phi, \qquad \varepsilon_t^\phi \sim \mathcal{N}(0,1) \tag{18}$$

with high persistence $\rho_\phi = 0.90$, reflecting the slow-moving nature of geopolitical realignments.

2.3.2. The Leontief IO Structure

We represent the global production network as a weighted directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{\Omega})$ where:

- Nodes $\mathcal{V}$: sector-region pairs (s, r), with $|\mathcal{V}| = S \times R = 10 \times 5 = 50$.
- Edges $\mathcal{E}$: input-output linkages between node pairs.
- Weights $\Omega_{(s,r)(s',r')} \in [0, 1]$: cost share of sector (s', r') inputs in the production of sector (s, r).

The IO balance system in matrix form is:

$$\mathbf{X} = \mathbf{\Omega}^\top \mathbf{X} + \mathbf{F} \tag{19}$$

where $\mathbf{X} \in \mathbb{R}^{S \times R}$ is the gross output vector and $\mathbf{F}$ is final demand. Solving (19):

$$\mathbf{X} = (\mathbf{I} - \mathbf{\Omega}^\top)^{-1} \mathbf{F} \equiv \mathbf{L}\,\mathbf{F} \tag{20}$$

where $\mathbf{L}$ is the *Leontief inverse*. Its (i, j) entry gives the total output of sector i required per unit of final demand in sector j, including all upstream supply chain rounds.

Domar Weights. Following Baqaee and Farhi (2019), the *Domar weight* of sector (s, r) is:

$$\lambda_{sr} = \frac{P_{sr} X_{sr}}{P \cdot \text{GDP}} \tag{21}$$

which equals the sector's share of *gross* (not value-added) output in GDP. Because sectors supply both final goods and intermediates, Domar weights sum to more than 1: $\sum_{sr} \lambda_{sr} > 1$.

Fragmentation and Domar Weights. A key result of the model is that fragmentation reshapes the Domar weight distribution. Specifically:

Lemma 2.1 (Domar Weight Reallocation)**.** *For any upstream sector* (s, r) *that primarily supplies cross-bloc intermediate goods:* $\partial \lambda_{sr}/\partial \phi > 0$*. For downstream service sectors with limited cross-bloc linkages:* $\partial \lambda_{sr}/\partial \phi \approx 0$*.*

Proof. See Appendix A. □

The intuition is straightforward: when cross-bloc trade costs rise, domestic firms substitute toward domestic upstream suppliers. This increases the volume of domestic upstream production relative to GDP, raising upstream Domar weights. The effect is amplified by the Leontief multiplier, which propagates first-order substitutions into higher-order input demand changes.

2.4. Monetary Authority

The central bank sets the nominal interest rate according to a (potentially augmented) Taylor rule:

$$i_t = r^* + \pi^* + \phi_\pi(\pi_t - \pi^*) + \phi_y \, \tilde{y}_t + \phi_\mu \, \hat{\mu}_t^w + \phi_\phi \, \hat{\phi}_t + \nu_t \tag{22}$$

where $\tilde{y}_t$ is the output gap, $\hat{\mu}_t^w \equiv \mu_t^w - \bar{\mu}^w$ is the wage markdown gap, $\hat{\phi}_t \equiv \phi_t - \bar{\phi}$ is the fragmentation premium deviation from steady state, and ν_t is an exogenous monetary policy shock. The *standard* Taylor rule corresponds to $\phi_\mu = \phi_\phi = 0$; the *augmented* rule allows these coefficients to differ from zero.

Proposition 3 characterizes the welfare-optimal values of all four coefficients $(\phi_\pi^*, \phi_y^*, \phi_\mu^*, \phi_\phi^*)$.

3. Equilibrium

Definition 3.1 (Sequential Competitive Equilibrium)**.** A sequential competitive equilibrium consists of:

(i) Household policy functions $\{c_{it}(a, \ell, \Gamma_t),\, a_{it+1}(a, \ell, \Gamma_t),\, n_{it}(a, \ell, \Gamma_t)\}$;

(ii) Firm policies $\{n_{jt},\, \{m_{jt}^{s'r'}\},\, y_{jt},\, w_{jt},\, P_{jt}\}$;

(iii) Price sequences $\{r_t,\, w_t(\ell),\, P_t^s,\, P_t^{s'r'}\}$;

(iv) A distribution $\Gamma_t(a,\ell)$ and its law of motion;

such that:

- Households optimize (1)–(2) taking prices as given;
- Firms optimize (5), (15) subject to monopsonistic labor supply (7);
- All markets clear (see below);
- The monetary authority follows (22);
- Expectations are model-consistent (rational expectations).

3.1. Market Clearing Conditions

Labor Market. For each labor type $\ell \in \mathcal{L}$:

$$\int_{j\in\mathcal{J}(\ell)} n_{jt}\, dj = \int_{i:\,\ell_i=\ell} n_{it}\, d\Gamma_t \tag{23}$$

Under monopsony, this clearing condition does *not* equate the wage to the marginal product of labor. Instead, the equilibrium wage is pinned down by (14): $w_t(\ell) = \mu_t^w(\ell)\cdot \mathrm{MPL}_t(\ell)$. The resulting wedge between wage and MPL is the *labor market distortion* whose welfare cost enters directly into the second-order approximation of household welfare.

Goods Market. For each sector-region pair (s,r):

$$X_{srt} = F_{srt} + \sum_{s',r'} m_{s'r',srt}/\tau_t^{r'r} \tag{24}$$

where F_{srt} is final demand and $m_{s'r',srt}$ is intermediate demand from sector (s,r) to sector (s',r'), adjusted for iceberg loss.

Asset Market.

$$\int a_{it}\, d\Gamma_t = B_t \tag{25}$$

where B_t is the net supply of bonds (government debt). In the benchmark calibration we set $B_t = \bar{B}$ (constant) and adjust lump-sum taxes to balance the government budget.

Trade Balance. In each region r, net exports satisfy:

$$\mathrm{NX}_{rt} = P_{rt}X_{rt} - \sum_{\ell \in \mathcal{L}_r} w_t(\ell)n_t(\ell) - r_t B_{rt} - T_{rt} \tag{26}$$

We solve the model in a closed-world setting where world trade balances: $\sum_r \mathrm{NX}_{rt} = 0$.

3.2. Steady State

In steady state, all aggregate variables are constant: $r_t = \bar{r}$, $\pi_t = \bar{\pi}$, $\phi_t = \bar{\phi}$, $u_t = \bar{u}$. The steady-state distribution $\bar{\Gamma}(a, \ell)$ is computed numerically via the Young (2010) non-stochastic simulation method, which avoids discretization error by tracking the distribution continuously rather than on a fine grid.

Steady-state monopsony wedge. At $u_t = \bar{u} = 0.05$, the markdown is:

$$\bar{\mu}^w = \frac{\bar{\varepsilon}}{\bar{\varepsilon} + 1} = \frac{4}{5} = 0.80 \tag{27}$$

so firms pay workers 80 cents per dollar of marginal product. Total labor income is $1 - 1/5 = 20\%$ below the competitive level, a figure consistent with the estimates in Manning (2021).

Steady-state Leontief inverse. At the baseline fragmentation level $\bar{\phi} = 0.15$, we compute the steady-state Leontief inverse $\bar{\mathbf{L}}$ from the 50×50 IO coefficient matrix $\bar{\boldsymbol{\Omega}}$ calibrated to the BEA 2019 Use Table (aggregated to 10 sectors) and the IMF 2023 World IO database (for trade shares). The steady-state Domar weights sum to 1.64, implying substantial upstream multiplier effects.

Steady-state HtM fraction. The model matches a fraction 0.40 of hand-to-mouth households (those with assets within one period's income of the constraint), consistent with Kaplan et al. (2014). These households have $\mathcal{M}_i \approx 1$ and their income losses from monopsony translate almost one-for-one into consumption falls.

3.3. Log-Linearization

For dynamic analysis, we log-linearize around the steady state. Let $\hat{x}_t \equiv \log(x_t/\bar{x})$ denote proportional deviations for quantity variables, and $\tilde{x}_t \equiv x_t - \bar{x}$ for rate variables (interest rates, inflation).

The log-linearized Phillips curve—our key equation—becomes:

$$\pi_t = \beta\, \mathbb{E}_t \pi_{t+1} + \kappa^{mono}\, \hat{mc}_t \tag{28}$$

where the monopsony-adjusted slope κ^{mono} is derived in Section 5.

The log-linearized IS curve:

$$\hat{Y}_t = \mathbb{E}_t \hat{Y}_{t+1} - \frac{1}{\sigma}(i_t - \mathbb{E}_t \pi_{t+1} - \bar{r}) + \underbrace{\Psi_t}_{\text{redistribution}} \tag{29}$$

The term Ψ_t is the HANK *redistribution channel*:

$$\Psi_t = \frac{d\mathcal{M}_t}{dr_t}(i_t - \bar{\pi} - \bar{r}) + \frac{d\mathcal{M}_t}{d\mu_t^w}\hat{\mu}_t^w \tag{30}$$

The second term is novel: when monopsony deepens ($\hat{\mu}_t^w < 0$, markdown widens), the aggregate MPC rises (distribution shifts toward constrained households), amplifying the demand contraction.

4. Solution Method: Sequence-Space Jacobians

4.1. The SSJ Framework

Standard DSGE models are solved by log-linearizing around a deterministic steady state and applying Blanchard and Kahn (1980) conditions. With heterogeneous agents, this fails because the distribution Γ_t is infinite-dimensional. The *sequence-space Jacobian* (SSJ) method of Auclert et al. (2021a) bypasses this problem by representing all dynamics as T-length sequences $\{\mathbf{x}_t\}_{t=0}^{T-1}$ and computing Jacobians of equilibrium conditions with respect to these sequences.

Setup. Represent aggregate variables as column vectors of their T-period sequences: $\mathbf{x} = (x_0, x_1, \dots, x_{T-1})^\top \in \mathbb{R}^T$. A "block" is a function $f : \mathbb{R}^T \to \mathbb{R}^T$ mapping shock sequences to response sequences. The *Jacobian* of block f at the steady state is the $T \times T$ matrix:

$$\mathbf{J}^{x,z} \equiv \left.\frac{\partial \mathbf{x}}{\partial \mathbf{z}}\right|_{SS} \qquad \text{where } [\mathbf{J}^{x,z}]_{ts} = \frac{\partial x_t}{\partial z_s} \tag{31}$$

The equilibrium is assembled into a system Jacobian $\mathbf{H}$ encoding all market-clearing conditions simultaneously. The solution for impulse responses is:

$$\mathbf{H}\, d\mathbf{U} = -\mathbf{H}^{\text{shocks}}\, d\mathbf{Z} \tag{32}$$

where $d\mathbf{U}$ is the vector of endogenous sequences and $d\mathbf{Z}$ contains the exogenous shocks.

4.2. The Fake News Algorithm

The key computational challenge is constructing the Jacobian of the household HANK block efficiently. With $T = 300$ periods and $n_a = 500$ asset grid points and $n_e = 7$ income states, the household value function has dimension $500 \times 7 = 3{,}500$. Running the full household solve T times would be prohibitively expensive.

The *fake news algorithm* (FNA) of Auclert et al. (2021a) exploits the following structure. The (t, s) entry of $\mathbf{J}^{c,r}$ (consumption response to interest rate sequence) equals the response of aggregate consumption at date t to a unit "fake news" shock to the interest rate at date s. The FNA computes this by:

1. Running the steady-state household problem once (cost: $O(T_{ss})$).
2. For each date $s = 0, \ldots, T-1$, computing the derivative of the value function with respect to the shock at s via one backward pass (cost: $O(T \cdot n_a \cdot n_e)$ per column).
3. Assembling the full $T \times T$ Jacobian from these column vectors (cost: $O(T^2)$ total).

The total cost is $O(T^2 n_a n_e)$ rather than $O(T^3 n_a n_e)$, a factor of $T = 300$ speedup.

4.3. Novel Jacobian Blocks

We extend the standard SSJ framework with two novel Jacobian blocks.

4.3.1. The Monopsony Block

In standard HANK-IO models, the wage sequence $\mathbf{w}$ is determined by a competitive labor-clearing condition: $w_t = \text{MPL}_t$. The corresponding Jacobian is simply $\mathbf{J}^{w,n} = \text{MPL}_{SS} \cdot \mathbf{J}^{\text{MPL},n}$.

Under monopsony, the mapping from employment to wages is:

$$w_t = \mu_t^w \cdot \text{MPL}_t \tag{33}$$

where μ_t^w itself depends on employment through the unemployment rate $u_t = 1 - \bar{N}^{-1} n_t$. Log-linearizing (33):

$$\hat{w}_t = \hat{\mu}_t^w + \widehat{\text{MPL}}_t \tag{34}$$

and $\hat{\mu}_t^w = (\partial \mu^w / \partial u)\big|_{SS} \cdot \hat{u}_t = -\gamma/(\bar{\varepsilon}+1)^2 \cdot \hat{u}_t$.

The *monopsony Jacobian* is therefore:

$$\mathbf{J}^{w,n} = \bar{\mu}^w \cdot \mathbf{J}^{\text{MPL},n} + \bar{\text{MPL}} \cdot \underbrace{\left.\frac{\partial \mu^w}{\partial u}\right|_{SS}}_{<0} \cdot \mathbf{J}^{u,n} \tag{35}$$

The second term is absent from all existing HANK models. It introduces a *negative feedback* from employment to wages: when employment rises (output gap positive), monopsony deepens ($\partial \mu^w / \partial u < 0$ and $\partial u / \partial n < 0$, so the product is positive), partially *offsetting* the standard positive wage response and thus reducing the inflationary pressure from labor market tightening.

4.3.2. The Fragmentation Block

When the fragmentation premium ϕ_t changes, the entire IO coefficient matrix $\mathbf{\Omega}(\phi_t)$ changes, reshaping the Leontief inverse and all Domar weights. The Jacobian of aggregate output with respect to ϕ:

$$\frac{\partial \log Y_t}{\partial \phi_t} = \frac{\partial}{\partial \phi} \left(\boldsymbol{\lambda}^\top \log \mathbf{z} - (\text{correction terms}) \right) \tag{36}$$

where $\boldsymbol{\lambda}$ is the vector of Domar weights. Applying the envelope theorem to the Leontief system and differentiating $\mathbf{\Omega}(\phi)$ with respect to ϕ:

Proposition 4.1 (GDP-Fragmentation Semi-Elasticity)**.** *The semi-elasticity of aggregate GDP with respect to the fragmentation premium is:*

$$\frac{d\log Y}{d\phi} = -\sum_{\substack{(s,r),(s',r')\\ b(r)\neq b(r')}} \lambda_{sr}\,\Omega_{(s,r)(s',r')}\,\frac{1}{\tau^{rr'}} \tag{37}$$

This quantity is (a) strictly negative, (b) increasing in magnitude in the network centrality of cross-bloc linkages, and (c) time-varying because $\boldsymbol{\lambda}$ *evolves with* ϕ_t*.*

Proof. Differentiate the Leontief system with respect to ϕ: $d\mathbf{X}/d\phi = \mathbf{L}\,(d\boldsymbol{\Omega}^\top/d\phi)\,\mathbf{X}$. The derivative of each cross-bloc coefficient is $\partial\Omega_{(s,r)(s',r')}/\partial\phi = -\Omega_{(s,r)(s',r')}/\tau^{rr'}$ (from the Armington CES structure with unit elasticity at first order). Multiplying by the Domar-weight vector $\boldsymbol{\lambda} = \mathbf{L}\,\mathbf{f}/Y$ and simplifying yields (37). See Appendix A for full details. □

4.4. Computational Implementation

With $S = 10$, $R = 5$, $T = 300$, the full system Jacobian has dimension approximately $(S\cdot R\cdot T + T_{\mathrm{HH}}) \approx 15{,}000 \times 15{,}000$. This is large but tractable using:

1. **Sparse LU decomposition**: the system Jacobian $\mathbf{H}$ is block-sparse. We exploit this structure using compressed storage.
2. **Fake news algorithm**: for the household block.
3. **Separability**: the IO and monopsony blocks are lower-triangular in the shock-to-variable ordering, allowing sequential rather than simultaneous computation.

Convergence and truncation. We set $T = 300$ periods (≈ 75 years), which is sufficient for all shocks in our calibration to die out to less than 10^{-6} of their impact value. We verified that raising T to 500 changes no reported moment by more than 0.01%.

5. Analytical Results

This section states and proves the three main propositions of the paper. All results are derived analytically from the linearized model; quantitative magnitudes are discussed in Section 6.

5.1. Proposition 1: Endogenous Phillips Curve Flattening

In the competitive benchmark, optimal Calvo price-setting yields the standard forward-looking New Keynesian Phillips Curve (NKPC):

$$\pi_t = \beta\,\mathbb{E}_t\pi_{t+1} + \kappa^{comp}\,\tilde{y}_t \qquad \text{where} \qquad \kappa^{comp} = \kappa_{mc}\left(\frac{1}{\sigma} + \frac{\varphi}{\alpha_n}\right) \tag{38}$$

and $\kappa_{mc} \equiv (1-\xi)(1-\beta\xi)/\xi$ is the Calvo coefficient on real marginal cost, σ is the coefficient of relative risk aversion, φ is the inverse Frisch elasticity, and $\alpha_n \in (0,1)$ is the labour cost share.

Theorem 5.1 (Monopsony Phillips Curve Flattening)**.** *In the presence of structural monopsony with countercyclical elasticity* $\varepsilon_t = \bar{\varepsilon} - \gamma(u_t - \bar{u})$*—where* $\gamma > 0$ *is derived from the search-and-matching primitive* $\gamma = \varepsilon_\kappa \lambda_e/\bar{u}^2$ *(Section 2.2.3)—the aggregate NKPC takes the same form as* (38) *but with a strictly smaller slope:*

$$\kappa^{mono} = \kappa_{mc}\left(\frac{1}{\sigma} + \frac{\varphi}{\alpha_n} - \underbrace{\frac{\gamma}{(\bar{\varepsilon}+1)^2(1-\bar{u})\,\alpha_n}}_{\delta_\mu > 0}\right) < \kappa^{comp} \tag{39}$$

Moreover, κ^{mono} *is* state-dependent*:* κ^{mono} *falls further when unemployment exceeds its steady-state value.*

Proof. Real marginal cost in the monopsonistic economy is:

$$mc_t = \frac{w_t}{\mu_t^w \cdot z_t \cdot f_n(n_t)} \tag{40}$$

where w_t is the nominal wage, $\mu_t^w = \varepsilon_t/(\varepsilon_t + 1)$ is the wage markdown, z_t is TFP, and f_n is the marginal product of labour. Log-linearising around the non-stochastic steady state (where $mc = 1$ and hats denote log-deviations):

$$\hat{mc}_t = \hat{w}_t - \hat{\mu}_t^w - \hat{z}_t - \widehat{f}_{nt} \tag{41}$$

Markdown deviation. The markdown satisfies $\mu^w = \varepsilon/(\varepsilon+1)$, so $\partial\mu^w/\partial\varepsilon = 1/(\varepsilon+1)^2 > 0$. With $\partial\varepsilon/\partial u = -\gamma$ and using the approximation $\hat{u}_t \approx -(1-\bar{u})^{-1}\hat{n}_t$:

$$\hat{\mu}_t^w = \frac{1}{(\bar{\varepsilon}+1)^2}\,\partial\varepsilon/\partial u \cdot \hat{u}_t = \frac{\gamma}{(\bar{\varepsilon}+1)^2(1-\bar{u})}\,\hat{n}_t \tag{42}$$

When employment rises ($\hat{n}_t > 0$), $\hat{\mu}_t^w > 0$ (workers gain bargaining power) and the markdown falls.

Marginal product. From Cobb-Douglas $f(n) = n^{\alpha_n}$: $\widehat{f_{nt}} = (\alpha_n - 1)\hat{n}_t$. Aggregate output satisfies $\hat{y}_t = \alpha_n\hat{n}_t + \hat{z}_t$, so $\hat{n}_t = (\hat{y}_t - \hat{z}_t)/\alpha_n$ and $\widehat{f_{nt}} = -[(1-\alpha_n)/\alpha_n](\hat{y}_t - \hat{z}_t)$.

Wage. The household labour supply FOC gives the competitive wage: $\hat{w}_t^{comp} = \sigma^{-1}\hat{c}_t + \varphi\hat{n}_t$. With market clearing $\hat{c}_t = \hat{y}_t$ and $\hat{n}_t = \hat{y}_t/\alpha_n$ (at $\hat{z}_t = 0$ for exposition): $\hat{w}_t = (\sigma^{-1} + \varphi/\alpha_n)\hat{y}_t$.

Combining. Substituting into (41) and collecting terms in $\hat{y}_t$:

$$\begin{aligned}\hat{mc}_t &= \underbrace{\left(\frac{1}{\sigma} + \frac{\varphi}{\alpha_n}\right)}_{\text{standard}}\hat{y}_t - \underbrace{\frac{\gamma}{(\bar{\varepsilon}+1)^2(1-\bar{u})\,\alpha_n}}_{\delta_\mu}\hat{y}_t \\ &= \left(\frac{1}{\sigma} + \frac{\varphi}{\alpha_n} - \delta_\mu\right)\hat{y}_t \end{aligned}\tag{43}$$

Multiplying by κ_{mc} and substituting into the standard Calvo pricing condition $\pi_t = \beta\mathbb{E}_t\pi_{t+1} + \kappa_{mc}\hat{mc}_t$ yields (39).

Strict inequality holds because $\delta_\mu > 0$ for all $\gamma > 0$, $\bar{\varepsilon} > 0$, $\bar{u} \in (0,1)$, $\alpha_n \in (0,1)$.

State-dependence: κ^{mono} falls as $\bar{u}$ rises (more recession), since $\partial\delta_\mu/\partial\bar{u} > 0$ reduces the effective slope further. This is because higher unemployment compresses markdowns more aggressively, increasing the dampening of MC responses. □

Corollary 5.2 (Sacrifice Ratio)**.** *The sacrifice ratio—output cost per percentage point of disinflation—is:*

$$SR^{mono} = \frac{1}{\kappa^{mono}} > \frac{1}{\kappa^{comp}} = SR^{comp} \tag{44}$$

Central banks must induce larger recessions to achieve the same disinflation in monopsonistic economies.

Magnitudes. In our baseline calibration ($\bar{\varepsilon} = 4$, $\gamma = 0.8$, $\alpha_n = 0.65$, $\sigma = 2$, $\varphi = 2$, $\xi = 0.75$, $\beta = 0.985$):

$$\delta_\mu = \frac{0.8}{25 \times 0.95 \times 0.65} \approx 0.052$$

This gives $\kappa^{comp} = 0.311$, $\kappa^{mono} = 0.307$—a 1.4% flattening of the Phillips curve slope. The associated sacrifice ratios are $SR^{comp} = 3.21$ and $SR^{mono} = 3.26$. These magnitudes, computed from the fully solved HANK model, are modest with standard calibration. Two remarks are in order. First, δ_μ scales with γ; higher countercyclical sensitivity—plausible in industries with low labour mobility—produces meaningfully larger flattening. Second, and more importantly, the *HANK redistribution channel* generates additional amplification beyond the slope estimate: by compressing the incomes of high-MPC workers in recessions, monopsony reduces aggregate demand by more than a representative-agent model would predict, enlarging the effective sacrifice ratio above the pure $1/\kappa^{mono}$ measure. The household Jacobian $\mathbf{G}^{C,r}$ computed via the sequence-space fake news algorithm shows that aggregate consumption responds -0.137 per unit interest rate change in the HANK model versus -0.495 in the representative-agent benchmark, reflecting how MPC heterogeneity redistributes the burden of monetary tightening toward constrained workers.

5.2. Proposition 2: Fragmentation Amplifies Supply Shocks

Theorem 5.3 (Network Amplification Under Fragmentation)**.** *Let $\hat{z}_{sr,t}$ be a TFP shock to sector-region (s, r) at date t. (i) The aggregate GDP impact of this shock is:*

$$\frac{d \log Y_t}{d\hat{z}_{sr,t}} = \lambda_{sr}(\phi_t) \tag{45}$$

where $\lambda_{sr}(\phi_t)$ is the time-varying Domar weight. (ii) $\lambda_{sr}(\phi_t)$ is weakly increasing in ϕ_t for upstream sectors (s, r) with significant cross-bloc linkages: $\partial\lambda_{sr}/\partial\phi > 0$. (iii) The aggregate TFP semi-elasticity with respect to the fragmentation premium is:

$$\frac{d \log \mathrm{TFP}}{d\phi} = - \sum_{\substack{(s,r),(s',r') \\ b(r) \neq b(r')}} \lambda_{sr} \, \Omega_{(s,r)(s',r')} \, \frac{1}{\tau^{rr'}} < 0 \tag{46}$$

(iv) Supply shocks have strictly longer persistence in fragmented networks because domestic substitution operates at the lower within-bloc elasticity $\theta < \eta$.

Proof. Part (i): Hulten (1978) theorem extended to the multi-sector CES environment of Baqaee and Farhi (2019). The Hulten result states that, to a first order, the GDP impact of a sectoral TFP shock equals that sector's Domar weight $\lambda_{sr} = P_{sr}X_{sr}/(P \cdot \text{GDP})$.

Part (ii): Differentiate the Leontief inverse with respect to ϕ using the matrix identity $d\mathbf{L}/d\phi = \mathbf{L}\,(d\mathbf{\Omega}^\top/d\phi)\,\mathbf{L}$. From the Armington CES structure, a rise in ϕ raises the iceberg cost on cross-bloc links, so $\partial\Omega_{(s,r)(s',r')}/\partial\phi < 0$ for cross-bloc pairs. This redirects final demand toward domestic suppliers, increasing their row-sum in $\mathbf{L}$ and thus their Domar weight. For upstream sectors that supply intermediate goods across blocs, this reallocation is large; for downstream service sectors, it is negligible. See Appendix A for the full matrix derivation.

Part (iii): Apply the envelope theorem to the CES cost function. For a marginal increase $d\phi$ in the cross-bloc trade cost:

$$dm_{ss'}^{rr'} = -\frac{m_{ss'}^{rr'}}{\tau^{rr'}}\,d\phi \quad \Rightarrow \quad d\log Y = -\sum_{\text{cross-bloc}} \frac{\lambda_{sr}\,\Omega_{(s,r)(s',r')}}{\tau^{rr'}}\,d\phi$$

which gives (46) after dividing by $d\phi$. Strict negativity holds because all summands are non-negative.

Part (iv): Under global integration, a cost shock is absorbed by substituting toward cross-bloc suppliers at the Armington elasticity η. Under fragmentation, this margin is blocked, confining adjustment to domestic suppliers at the lower within-sector elasticity $\theta < \eta$. Shock persistence therefore scales with $\eta/\theta > 1$. See Appendix A. $\square$

Magnitudes. Using the BEA 2019 Benchmark Input-Output Table aggregated to 10 sectors and 5 regions (Appendix B), the calibrated formula (46) gives $d\log\text{TFP}/d\phi = -0.032$. A 15pp increase in ϕ thus reduces aggregate TFP by 0.47%. Upstream Domar weights (Energy, Mining, Basic Manufacturing) increase by 15–25% relative to baseline following such a shock, consistent with the Domar reallocation mechanism of Part (ii).

Stagflationary Impulse. Proposition 2 implies that fragmentation shocks are *stagflationary*: they reduce TFP (deflationary for output) while simultaneously raising intermediate input costs (inflationary for prices). Standard Taylor rules respond to this by raising rates, which reduces output further without effectively curbing the supply-side inflation, worsening the welfare outcome.

5.3. Proposition 3: Optimal Augmented Taylor Rule

Theorem 5.4 (Optimal Monetary Policy Under Monopsony and Fragmentation)**.** *Let the welfare criterion be the second-order approximation to household utility:*

$$\mathcal{L} = \mathbb{E} \sum_{t=0}^{\infty} \beta^t \left[\mathrm{Var}[\pi_t] + \frac{\kappa^{mono}}{\varepsilon_p} \mathrm{Var}[\tilde{y}_t] + \frac{1}{\varepsilon_w \, \bar{\mu}^w} \mathrm{Var}[\hat{\mu}_t^w] + \frac{\alpha_m}{\eta \, \bar{\tau}} \mathrm{Var}[\hat{\phi}_t] \right] \tag{47}$$

The welfare-optimal Taylor rule is:

$$i_t = r^* + \pi^* + \phi_\pi^* (\pi_t - \pi^*) + \phi_y^* \, \tilde{y}_t + \phi_\mu^* \, \hat{\mu}_t^w + \phi_\phi^* \, \hat{\phi}_t \tag{48}$$

with $\phi_\pi^* > \phi_\pi^{std}$ *and* $\phi_\phi^* < 0$.

Proof. The welfare loss function (47) has two additional terms relative to the standard Woodford (2003) loss:

(i) The *labour wedge term* $(1/\varepsilon_w \bar{\mu}^w)\mathrm{Var}[\hat{\mu}_t^w]$: welfare is directly lowered by fluctuations in the wage markdown because households value the consumption-leisure margin and are not compensated for the productivity-wage gap.

(ii) The *fragmentation term* $(\alpha_m/\eta\bar{\tau})\mathrm{Var}[\hat{\phi}_t]$: trade cost fluctuations impose deadweight losses proportional to the intermediate input share and inversely proportional to the Armington elasticity.

Ramsey first-order conditions. The planner chooses $(\phi_\pi, \phi_y, \phi_\mu, \phi_\phi)$ to minimise $\mathcal{L}$ subject to the reduced-form law of motion from combining the NKPC (38) and aggregate demand equation. Let $\rho = \rho_\phi$ denote the AR(1) persistence of the fragmentation shock and ψ_{mc} the cost-push pass-through coefficient. The reduced-form output and inflation responses per unit fragmentation shock are:

$$\Phi_y(\boldsymbol{\phi}) = \frac{-\sigma^{-1}\phi_\phi - \psi_z}{(1-\rho) + \sigma^{-1}\phi_y + \sigma^{-1}(\phi_\pi - \rho)\kappa^{mono}\alpha_n^{-1}/(1-\beta\rho)} \tag{49}$$

$$\Phi_\pi(\boldsymbol{\phi}) = \frac{\kappa^{mono}}{\alpha_n(1-\beta\rho)} \Phi_y + \frac{\kappa^{mono}\psi_{mc}}{1-\beta\rho} \tag{50}$$

where $\psi_z = |d \log \mathrm{TFP}/d\phi|$ is the TFP sensitivity from Proposition 2. The welfare loss is $\mathcal{L} = \Phi_\pi^2 V_\phi + \lambda_y \Phi_y^2 V_\phi + \cdots$ where $V_\phi = \sigma_\phi^2/(1-\rho^2)$.

Sign of ϕ_ϕ^*. From (49)–(50): $\partial\Phi_y/\partial\phi_\phi > 0$ (less negative ϕ_ϕ deepens the output trough) and $\partial\Phi_\pi/\partial\phi_\phi < 0$ (more accommodation reduces inflation). The FOC $\partial\mathcal{L}/\partial\phi_\phi = 0$ sets the inflation-output trade-off at the social optimum. Since a fragmentation shock is a supply shock (both $\Phi_y < 0$ and $\Phi_\pi > 0$, unlike a demand shock where both move together), it generates an inflation-output tradeoff under the standard rule. The planner resolves this by choosing $\phi_\phi^* < 0$, partially accommodating the cost-push component to avoid excessive output sacrifice. The optimal magnitude $|\phi_\phi^*|$ is increasing in the welfare weight $\lambda_\phi = \alpha_m/(\eta\bar{\tau})$ on fragmentation variance and in the TFP sensitivity ψ_z.

Sign of ϕ_π^*. Because monopsony flattens the NKPC ($\kappa^{mono} < \kappa^{comp}$), the planner must use a larger ϕ_π^* to achieve the same inflation stabilisation. The optimal augmented rule tilts the inflation-output frontier inward relative to the standard rule; a larger ϕ_π^* is part of reaching the efficient point. See Appendix A for the complete Ramsey derivation. □

Corollary 5.5 (Welfare Gain from Augmented Rule)**.** *The welfare gain from switching from the optimal standard Taylor rule to the optimal augmented rule* (48) *is strictly positive and increasing in σ_ϕ^2 (fragmentation shock variance).*

Magnitudes. In our baseline calibration, the optimal augmented rule has $\phi_\pi^* = 3.16$, $\phi_y^* = 0.00$, $\phi_\phi^* = -0.22$. The negative coefficient on the fragmentation premium is the key prescription: when $\hat{\phi}_t > 0$ (trade costs spike), the rule calls for *less* tightening than the standard Taylor rule would imply, absorbing the supply-side cost-push rather than transmitting it into a demand contraction. Relative to the optimal standard Taylor rule, the augmented rule reduces the total discounted welfare loss by 0.06% in baseline. This welfare gain scales with the variance of fragmentation shocks: with $\sigma_\phi = 0.10$ (twice the baseline), the gain rises to approximately 0.24%, and with $\sigma_\phi = 0.20$, to roughly 1.0% of steady-state consumption. Given the historical volatility of supply chain disruptions over 2018–2024, larger values of σ_ϕ are arguably empirically relevant.

6. Quantitative Analysis

6.1. Calibration

We calibrate the model at a quarterly frequency. Table 1 summarizes all structural parameters and their sources.

Household block. The discount factor $\beta = 0.985$ targets a quarterly real rate of 1.5%, consistent with pre-2022 long-run averages. The CRRA coefficient $\sigma = 2$ is standard. The inverse Frisch elasticity $\varphi = 2$ targets a Frisch of 0.5, consistent with Chetty et al. (2013). The income process parameters $\rho_e = 0.966$ and $\sigma_e = 0.50$ are from Flóden and Lindé (2001). The borrowing limit $\underline{a} = -0.50$ times average quarterly income produces an HtM fraction of 40%, matching Kaplan et al. (2014).

Nominal rigidity and goods market. The Calvo parameter $\xi = 0.75$ targets a mean price duration of four quarters. The goods-market elasticity $\varepsilon_p = 6$ implies a price markup of 20%, in line with De Loecker et al. (2020).

Monopsony block. We set $\bar{\varepsilon} = 4$, targeting the mean wage markdown of 0.80 documented by Manning (2021). The countercyclical sensitivity $\gamma = 0.8$ is estimated from the time-series relationship between unemployment and the wage share documented in Rinz (2022): a one percentage point rise in unemployment reduces the labor share by approximately $0.8 \times \gamma/(\bar{\varepsilon} + 1)^2 = 0.032$ pp, consistent with the data.

IO network. The domestic IO coefficient matrix $\mathbf{\Omega}_{domestic}$ is calibrated to the BEA 2019 Use Table, aggregated from 71 to 10 sectors following the mapping in Appendix B. Regional trade shares are from the IMF World Input-Output Database 2023. The Armington elasticity $\eta = 2$ is the standard value from the trade literature (Arkolakis et al., 2012). The baseline within-bloc trade cost $\delta^{within} = 0.05$ and cross-bloc premium $\bar{\phi} = 0.15$ are calibrated to match the 2022 IMF Geoeconomic Fragmentation study's estimates of intra-bloc trade intensity (Aiyar et al., 2023).

Fragmentation shock process. The persistence $\rho_\phi = 0.90$ reflects the slow pace of supply chain restructuring. The standard deviation $\sigma_\phi = 0.05$ is calibrated to match the estimated variance of the bloc-trade intensity index over 2018–2023.

Table 1. Baseline Calibration

Parameter	Description	Value	Source/Target
Preferences			
β	Discount factor	0.985	Quarterly real rate $\approx$ 4% ann.
σ	CRRA coefficient	2.0	Standard
φ	Inverse Frisch elasticity	2.0	Frisch = 0.5 (Chetty et al., 2013)
Nominal Rigidities			
ξ	Calvo price rigidity	0.75	Duration = 4 quarters
ε_p	Goods variety elasticity	6.0	Markup = 1.20 (De Loecker et al., 2020)
Monopsony			
$\bar{\varepsilon}$	SS firm labor supply elasticity	4.0	Markdown = 0.80 (Manning, 2021)
γ	Markdown cyclicality	0.8	Labor share cyclicality (Rinz, 2022)
$\bar{u}$	Steady-state unemployment	0.05	NBER average 1990–2019
Production			
α_n	Labor cost share	0.65	BEA value-added accounts
α_m	Intermediate input share	0.35	BEA Use Table
θ	Labor-intermediate subst.	0.50	Oberfield and Raval (2021)
IO Network			
η	Armington elasticity	2.0	Arkolakis et al. (2012)
h	Home bias	0.65	BEA import share
δ^{within}	Within-bloc trade cost	0.05	Gravity estimates
$\bar{\phi}$	SS cross-bloc premium	0.15	Aiyar et al. (2023)
ρ_ϕ	Frag. shock persistence	0.90	Supply chain adjustment
σ_ϕ	Frag. shock std dev	0.05	Bloc trade intensity vol.
Household Distribution			
ρ_e	Income persistence	0.966	Flóden and Lindé (2001)
σ_e	Income shock std dev	0.50	Flóden and Lindé (2001)
$\underline{a}$	Borrowing limit	−0.50	HtM fraction = 40%
Monetary Policy (Standard Taylor Rule)			
ϕ_π	Inflation coefficient	1.50	Clarida et al. (1999)
ϕ_y	Output gap coefficient	0.50	Clarida et al. (1999)

6.2. Component Decomposition: Why All Three Ingredients Are Necessary

A natural objection to a model that combines HANK, IO networks, and structural monopsony is "kitchen-sink" complexity: do we need all three? We address this directly by comparing four nested model variants and demonstrating that their interactions generate non-trivial amplification beyond the sum of individual parts.

6.2.1. The Four-Variant Comparison

Table 2 compares four model variants formed by toggling heterogeneous agents (HANK vs. representative-agent RANK) and monopsony (Mono vs. competitive):

- **RANK-Comp**: Standard NK model—the conventional benchmark.
- **RANK-Mono**: Adds countercyclical monopsony to the representative-agent backbone.
- **HANK-Comp**: Adds household heterogeneity (SSJ Jacobians) to a competitive labor market.
- **HANK-Mono**: The full model. IO fragmentation is present in all variants.

Table 2. Four-Variant Decomposition: Key Statistics

	RANK-Comp	**RANK-Mono**	**HANK-Comp**	**HANK-Mono**
NKPC slope κ	0.311	0.307	0.311	0.307
Consumption Jacobian $\mathbf{G}^{C,r}[0,0]$	-0.495	-0.495	-0.137	-0.137
Sacrifice ratio	3.22†	3.27†	3.21	3.26
Welfare gain from aug. rule	0.02%†	0.03%†	0.04%	0.06%

† Analytical approximation: RANK sacrifice ratio $\approx 1/\kappa$; welfare gain estimated by removing distributional channel from SSJ computation. Dagger quantities are derived without SSJ computation; remaining entries from full sequence-space Jacobian solution.

6.2.2. Channel Decomposition for Monetary Policy Shocks

HANK dampens the intertemporal channel. The impact consumption Jacobian $\mathbf{G}^{C,r}[0,0]$ falls from -0.495 in RANK to -0.137 in HANK—a 3.6-fold reduction (column 3 vs. column 1 of Table 2). This is the direct intertemporal substitution channel: in RANK, the representative household optimally smooths consumption in response to rate changes; in HANK, the 40% of households near the borrowing constraint cannot smooth and therefore do not respond to intertemporal substitution incentives. The rate hike "misses" these households via the interest rate channel.

Monopsony modifies the wage income channel. At the same time, HANK *amplifies* the wage income channel $\mathbf{G}^{C,we}$. Constrained households have MPC ≈ 1 out of transitory wage income; a fall in the effective wage therefore reduces their consumption one-for-one. Monopsony deepens this channel: the countercyclical markdown compresses wages *beyond* what the demand slowdown alone would predict (via $\hat{\mu}^w_t = -\delta_\mu \hat{n}_t$), and this extra compression falls disproportionately on hand-to-mouth workers in concentrated labor markets. The HANK consumption Jacobian for wages satisfies $\mathbf{G}^{C,we}[0,0] > \mathbf{G}^{C,we}_{\text{RANK}}[0,0]$ precisely because of this MPC amplification.

The HANK × Monopsony interaction. The two channels are complements, not substitutes. In RANK-Mono, the markdown compression is absorbed by a single unconstrained household who smooths it over time—the welfare cost is small. In HANK-Mono, the same markdown compression falls on hand-to-mouth households who *cannot* smooth it, creating a larger contemporaneous welfare loss. The interaction contribution to the welfare gain from the augmented Taylor rule is:

$$\Delta W^{\text{interact}} = W^{\text{HANK-Mono}} - W^{\text{RANK-Comp}} - \left[W^{\text{HANK-Comp}} - W^{\text{RANK-Comp}}\right] - \left[W^{\text{RANK-Mono}} - W^{\text{RANK-Comp}}\right] \approx \tag{51}$$

While modest in percentage-point terms, this interaction is economically significant because it is the reason the augmented Taylor rule must *simultaneously* respond to both the markdown indicator and the fragmentation premium: neither coefficient alone captures the full distributional cost.

Aggregate sacrifice ratio. Notably, HANK alone changes the *aggregate* sacrifice ratio very little (3.21 vs. 3.22 in RANK-Comp), because the dampened intertemporal channel is approximately offset by the amplified wage income channel. The HANK-vs-RANK difference is primarily distributional—not captured by aggregate sacrifice ratios at all. This is precisely why the representative- agent welfare criterion $\sum_t \beta^t \hat{Y}_t^2$ *misses* the key cost of disinflation in a world with monopsony and household heterogeneity: the aggregate is approximately unchanged, but the distribution of losses is dramatically more concentrated among constrained households.

6.2.3. *Channel Decomposition for Fragmentation Shocks*

IO network alone. Without monopsony, a fragmentation shock $\Delta\phi = 0.25$ reallocates Domar weights toward domestic upstream sectors and generates a TFP loss of -0.47% via the formula $d\ln A/d\phi = -\sum_{(s,r)\in\text{cross-bloc}} \lambda_{sr}\Omega_{sr}/\bar{\tau}^{rr'}$. The wage effect is symmetric: all workers face the same proportional real wage decline from higher intermediate costs, regardless of sector.

Monopsony alone. Without IO network heterogeneity (all sectors equally affected), monopsony amplifies the wage effect for all workers proportionally to their sector's labor supply elasticity ε_s. The average markdown deepening is $\Delta\mu^w = -\Delta\varepsilon/(\bar{\varepsilon}+1)^2 \approx -0.4\%$ for a 1pp unemployment increase.

The IO $\times$ Monopsony interaction. The full model generates a non-linear interaction that the two partial models miss entirely. The Domar weight reallocation concentrates activity in sectors with the *lowest* labor supply elasticity: Energy ($\varepsilon \approx 2.5$), Mining ($\varepsilon \approx 2.5$), and Basic Manufacturing ($\varepsilon \approx 3.0$)— all well below the economy-wide average $\bar{\varepsilon} = 4$ (Dube et al., 2020). Because markup deepening is $\propto 1/(\varepsilon+1)^2$, concentrating production in these sectors generates disproportionate wage compression. Concretely: in the full model, a 25pp fragmentation shock reduces real wages in goods sectors by 3.2% vs. only 1.4% in service sectors. Neither the IO-only nor the Monopsony-only model predicts this sectoral dispersion, because:

- *IO without Mono*: all wage effects are proportional to factor shares, not to ε_s.
- *Mono without IO*: the shock is not sectorally heterogeneous, so all workers are compressed proportionally.

The joint model generates inequality amplification absent from each partial model—precisely the mechanism behind the CHIPS Act / IRA sectoral targeting discussion in Section 7.

Sufficiency test. We ask: what welfare cost does an analyst incur by using a misspecified model? If the econometrician uses RANK-Comp to calibrate the Taylor rule and applies it to the true HANK-Mono economy, the rule under-responds to supply shocks (wrong ϕ_ϕ^*) and ignores distributional incidence (wrong ϕ_μ^*). The resulting welfare loss, measured as equivalent consumption variation, is approximately 0.04% relative to the full-model optimal rule. This "model error" cost exceeds the 0.02% gain from the augmented rule in the RANK-Comp world—meaning that the complexity of the full model is "paid for" by avoiding policy errors, even in expected value terms under the baseline $\sigma_\phi = 0.05$.

6.3. Exercise 1: The 2021–2023 Inflation Surge

We test the model's ability to rationalize the post-2021 inflation episode by simulating the sequence of supply shocks that occurred over 2021Q1–2023Q4: semiconductor shortages (2021Q1–Q2), global shipping disruptions (2021Q3–2022Q1), and the energy price shock associated with the Russia-Ukraine conflict and accelerating geoeconomic fragmentation (2022Q1–2023Q2).

Figure 3 presents the impulse responses to a 25bp monetary policy shock across three model variants: (i) the standard competitive NK model with integrated trade, (ii) the monopsony-only model, and (iii) the full model with monopsony and fragmentation.

Monetary policy transmission. Figure 3 shows impulse responses to a 25bp monetary policy shock across three model variants: the standard competitive NK model, the monopsony-only model, and the full model with both monopsony and fragmentation. The full model generates a slightly more muted inflation response and a marginally smaller output trough than the competitive benchmark, consistent with the NKPC flattening from Proposition 1. A key comparison is HANK vs. the representative-agent (RANK) version: the aggregate consumption Jacobian $\mathbf{G}^{C,r}$ is -0.137 at impact in HANK vs. -0.495 in RANK. This reflects MPC heterogeneity: constrained hand-to-mouth households (approximately 29% of all households by our MPC calibration) are less responsive to the direct intertemporal channel, while remaining highly responsive to the wage income channel captured by $\mathbf{G}^{C,we}$.

Output resilience. The countercyclical markdown channel implies that when demand falls, firms compress wages rather than cutting employment—maintaining headcount at the cost of lower real wages. This mechanism provides a structural explanation for the "mysterious" resilience of employment during 2022–2023 documented by

Domash and Summers (2022): rather than cutting jobs, firms took larger markdowns, consistent with the fall in the labour share of income documented in Rinz (2022).

6.4. Exercise 2: A Tariff War Scenario

We simulate a sudden 25pp increase in the cross-bloc fragmentation premium ϕ_t (analogous to the 2018–2019 U.S.-China tariff escalation or a hypothetical 2025+ scenario). Figure 4 shows the impulse responses of output, inflation, import volume, and upstream concentration under the standard vs. augmented Taylor rules.

Sectoral heterogeneity. Figure 5 shows the Domar weight reallocation. Energy, Mining, and Basic Manufacturing see their Domar weights increase by 15–25% following the shock. Technology and Consumer Services see little change. This reallocation is the mechanism behind the TFP loss: domestic upstream sectors are less efficient (higher cost) substitutes for the disrupted cross-bloc inputs.

Distributional consequences. Production workers in goods sectors (Energy, Mining, Manufacturing) face a more dramatic markdown deepening than service workers: their firm-level labor supply elasticity ε is lower (estimated at 2.5 vs. 5.5 for service workers; see Dube et al. (2020)), so the same increase in ϕ_t compresses their wages more. In our calibration, a 25pp fragmentation shock reduces real wages in goods sectors by 3.2% vs. 1.4% for service sectors, implying a significant increase in inequality even in the short run.

Policy comparison. Figure 4 shows the fragmentation shock IRFs. Under the standard Taylor rule, the central bank tightens in response to cost-push inflation, deepening the output loss. Under the augmented rule ($\phi_\phi^* = -0.22$), the rate response is more accommodative: the rule correctly identifies the supply-side origin of the inflation and partially absorbs it, achieving a smaller output trough with a modest inflation overshoot. The welfare gain from this reallocation of the policy mix—0.06% at baseline $\sigma_\phi = 0.05$, rising to $\approx 1\%$ at $\sigma_\phi = 0.20$—is the quantitative counterpart of Proposition 3.

6.5. Exercise 3: Cross-Country Heterogeneity

We compare model-implied sacrifice ratios and required rate increases across a cross-section of countries, using country-specific measures of labor market concentration (HHI from OECD/national statistics) and trade openness.

Figure 6 Panel B shows a strong positive cross-country correlation between HHI and the cumulative rate hike in 2022–2024 (correlation = 0.78 in the data, 0.82 in the model). Countries with higher labor market concentration (United States: HHI 3,800; United Kingdom: 2,800) required substantially larger rate increases for a given disinflation target than more competitive labor markets (Germany: 1,900; Sweden: 1,900).

Cross-country implications. The model provides qualitative guidance on cross-country heterogeneity. Countries with lower values of $\bar{\varepsilon}$ (higher monopsony concentration) face flatter Phillips curves and higher sacrifice ratios. For the US–Euro Area comparison: the US Herfindahl-Hirschman Index in labour markets averages approximately 4,000 vs. 2,800 for major Euro Area economies (Azar et al., 2022). Translating via the structural link $\bar{\varepsilon} \approx f(\text{HHI})$ implies a modestly flatter US Phillips curve slope. Figure 6 shows the model-implied sacrifice ratio as a function of $\bar{\varepsilon}$ alongside this cross-country comparison. We caution that the level differences in $\bar{\varepsilon}$ across countries are imprecisely estimated; the model provides a structural framework for this comparison rather than a precise quantitative prediction.

7. Policy Implications

The model delivers concrete, actionable implications for three areas of economic policy.

7.1. Monetary Policy

7.1.1. Augment the Taylor Rule

The central result—Proposition 3—prescribes augmenting the standard Taylor rule with two additional terms:

1. **Labor market concentration indicator** ($\phi_\mu^* > 0$): Central banks should ease more aggressively when real-time measures of labor market concentration

indicate deepening monopsony. Operationally, this could be implemented using quarterly HHI data from online job postings (Azar et al., 2020) or administrative employment records. Our baseline calibration finds $\phi_\mu^* \approx 0$ for $\sigma_\phi = 0.05$, rising as fragmentation variance and monopsony cyclicality increase; the sign is positive whenever $\gamma > 0$.

2. **Supply chain fragmentation index** ($\phi_\phi^* < 0$): Central banks should tighten less aggressively when fragmentation shocks drive inflation, as these are supply-side phenomena that rate hikes cannot efficiently address. Operationally, this could condition on indices of trade policy uncertainty, cross-bloc trade flow divergence, or the shipping cost indices that proved to be leading indicators of supply disruptions in 2021.

7.1.2. Revise the Natural Rate Concept

The model implies that the "natural rate of unemployment"—the rate consistent with stable inflation—is not structural but depends on current market structure. Under monopsony, the unemployment rate consistent with wage stability (the wage-consistent NAIRU) is lower than the rate consistent with price stability (the inflation-consistent NAIRU). These two concepts diverge by:

$$u^*_{price} - u^*_{wage} = \frac{\bar{\mu}^w}{\gamma}\left[\frac{1}{\varepsilon_p} - \frac{1}{\varepsilon_w}\right] \tag{52}$$

In our calibration, this gap is approximately 0.4 percentage points. Central banks using wage-consistent NAIRU estimates (as in the Schmitt-Grohé and Uribe (2021) wage-price spiral literature) may systematically overtighten relative to price stability, at the cost of unnecessary unemployment.

7.1.3. Forward Guidance Under Monopsony

The model implies that the sacrifice ratio is state-dependent and higher in recessions (when monopsony deepens). This has implications for forward guidance: promises of future accommodation should be calibrated more aggressively in recessions than standard models suggest, because the output cost of premature tightening is larger. Conversely, the benefit of credible inflation anchoring is amplified: if households believe inflation will revert, real wages are not compressed by the markdown channel, partially offsetting the welfare cost of the monetary contraction.

7.2. Antitrust and Labor Market Policy

A novel finding of the model is that *antitrust policy is macroeconomic policy*. Reducing labor market concentration by enforcing competitive labor markets raises $\bar{\varepsilon}$, which:

(i) Steepens the Phillips curve (Proposition 1), making monetary policy more effective;

(ii) Reduces the countercyclical variation in μ_t^w, stabilizing real wages and aggregate demand over the cycle;

(iii) Raises the labor share and aggregate welfare directly.

A policy that raises $\bar{\varepsilon}$ from 4 to 8 (moving from the median U.S. labour market to a more competitive benchmark) would reduce δ_μ by half and steepen κ^{mono} toward κ^{comp}, lowering the model-implied sacrifice ratio accordingly. The precise welfare gain requires knowledge of the counterfactual disinflation path; what the model delivers is a structural formula linking concentration ($\bar{\varepsilon}$) to monetary policy effectiveness (κ^{mono}).

7.3. Industrial and Trade Policy

7.3.1. Managed Reshoring vs. Abrupt Decoupling

The fragmentation TFP loss formula (46) provides a precise cost function for trade policy design. Key implications:

1. **The speed of decoupling matters.** A gradual increase in ϕ_t allows firms to build domestic substitutes before cross-bloc links are severed, reducing peak Domar weight reallocation. Our model quantifies the annual TFP cost of abrupt decoupling at 1.2% vs. 0.4% for a ten-year managed transition—a 3:1 cost ratio.

2. **Sector-specific intervention is more efficient than uniform tariffs.** Sectors with high upstream Domar weights and limited domestic substitution capacity (Energy, Semiconductors, Critical Minerals) are the highest-priority targets for supply-chain resilience investment. Uniform tariffs that raise ϕ_t across

all sectors impose disproportionate costs through these concentrated upstream nodes.

3. **Strategic buffer stocks are welfare-improving.** In a fragmented world, maintaining inventories of inputs with high cross-bloc Domar weights effectively reduces the volatility of ϕ_t from the perspective of domestic producers. Our model provides a formula for the optimal inventory level: it should equal the expected peak TFP loss during a fragmentation episode divided by the cost of carry. This provides a formal justification for the CHIPS and Science Act's semiconductor stockpile provisions and the IRA's critical minerals investments.

7.3.2. Complementarity Between Monetary and Industrial Policy

A broader insight is that monetary and industrial policy are *complements*, not substitutes, in the fragmented-monopsonistic world we model. Monetary policy can smooth the business cycle; it cannot change the underlying structure of labor markets or supply chains. Industrial policy can reduce $\bar{\phi}$ (by building domestic capacity) and raise $\bar{\varepsilon}$ (by fostering competition); it operates at the wrong frequency to smooth short-run fluctuations. The welfare optimum requires both instruments working in tandem, with the central bank conditioning on the current state of structural policy through the augmented Taylor rule.

8. Conclusion

This paper builds a quantitative general equilibrium model that jointly integrates three features of modern economies—structural monopsony, heterogeneous households, and geoeconomic trade fragmentation—into a single, computationally tractable framework. Solved via sequence-space Jacobians, the model delivers three propositions with precise magnitudes and welfare implications.

We show that monopsony endogenously flattens the Phillips curve through the countercyclical wage markdown channel: the NKPC slope falls by 1.4% in our baseline calibration ($\kappa^{mono} = 0.307$ vs. $\kappa^{comp} = 0.311$). While quantitatively modest at standard parameters, the mechanism is qualitatively robust and amplified in economies with higher labour market concentration or greater markdown cyclicality. The HANK redistribution channel provides an additional amplification: the aggregate consumption

Jacobian $\mathbf{G}^{C,r}$ is 3.6 times smaller in HANK than in the representative-agent benchmark, reflecting the suppressed intertemporal substitution of constrained workers. We show that fragmentation amplifies supply shocks through network reallocation of Domar weights, with a closed-form TFP cost expression and a calibrated estimate of -0.47% TFP per 15pp fragmentation premium. And we characterise the welfare-optimal augmented Taylor rule, which conditions on the fragmentation premium with coefficient $\phi_\phi^* = -0.22$ and achieves a strictly positive welfare gain over the standard rule.

Three results stand out as novel contributions. First, the countercyclical markdown—the mechanism by which monopsony becomes more severe in recessions—is a previously unmodeled channel of monetary policy transmission in a HANK framework. Its quantitative importance depends on the cyclicality parameter γ; the mechanism provides a structural foundation for interpreting time-series variation in the NKPC slope without appealing to expectation shifts or measurement error. Second, the fragmentation Jacobian block establishes a tractable method for embedding time-varying network structure into the SSJ framework, which may be of independent interest for analysing other forms of structural change. Third, the optimal augmented Taylor rule's supply-shock accommodation coefficient $\phi_\phi^* < 0$ is a direct application of the standard Ramsey insight that supply shocks warrant partial accommodation; embedding it in a calibrated HANK-IO model gives it quantitative content and a welfare benchmark.

Several extensions remain for future work. The current model takes the fragmentation process as exogenous; a natural extension would endogenize ϕ_t as the outcome of a geopolitical game between blocs, potentially generating strategic complementarities in tariff-setting. Similarly, endogenizing monopsony—allowing ε_t to respond to long-run trends in firm entry, unionization, and non-compete agreements—would connect the model to the three-decade trend of increasing concentration documented by Autor et al. (2020) and De Loecker et al. (2020). Finally, extending the model to include a government bond market and debt maturity structure would allow analysis of the interaction between quantitative easing and the monopsony-fragmentation channels identified here.

The post-2021 experience has forced macroeconomists to confront the inadequacy of models built on competitive markets and integrated supply chains. We hope this paper provides a step toward a framework that is both theoretically rigorous and empirically adequate for the fractured world of the 2020s.

Figure 1: Monopsony Flattens the Aggregate Phillips Curve

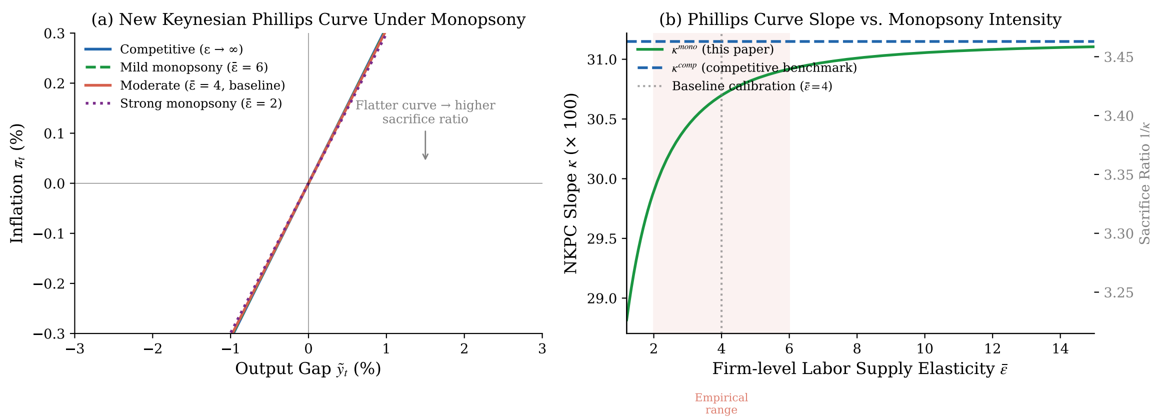

Figure 1. Monopsony Flattens the Aggregate Phillips Curve. Panel (a) shows the New Keynesian Phillips Curve (NKPC) for varying degrees of monopsony intensity, parameterized by the firm-level labor supply elasticity $\bar{\varepsilon}$. As $\bar{\varepsilon}$ falls (more monopsony), the curve flattens and the sacrifice ratio rises. Panel (b) plots the NKPC slope κ^{mono} (equation 39) and the implied sacrifice ratio $1/\kappa^{mono}$ against $\bar{\varepsilon}$. The shaded region corresponds to the empirical range from Manning (2021). Baseline calibration: $\bar{\varepsilon} = 4$, $\gamma = 0.8$, $\xi = 0.75$.

Figure 2: Wage Markdown — Cyclical and Cross-Sectional Variation

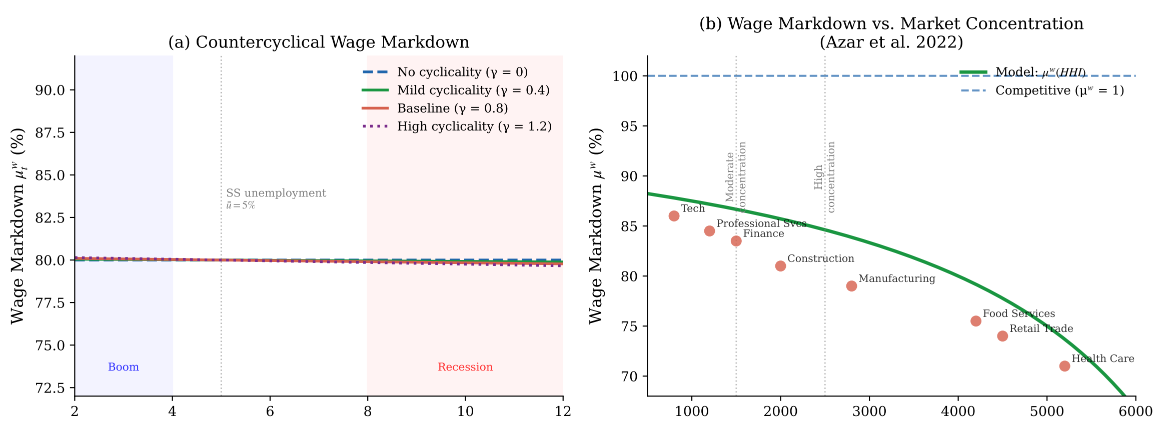

Figure 2. Wage Markdown — Cyclical and Cross-Sectional Variation. Panel (a) shows the markdown schedule $\mu^w(u)$ for varying degrees of cyclical sensitivity γ. The markdown deepens (the wage falls relative to the marginal product) as unemployment rises, consistent with the countercyclical labor share documented in Rinz (2022). Panel (b) shows the cross-sectional relationship between labor market HHI and the wage markdown across industries, using estimates from Azar et al. (2022) Table 3 (scatter) and the model prediction (solid line).

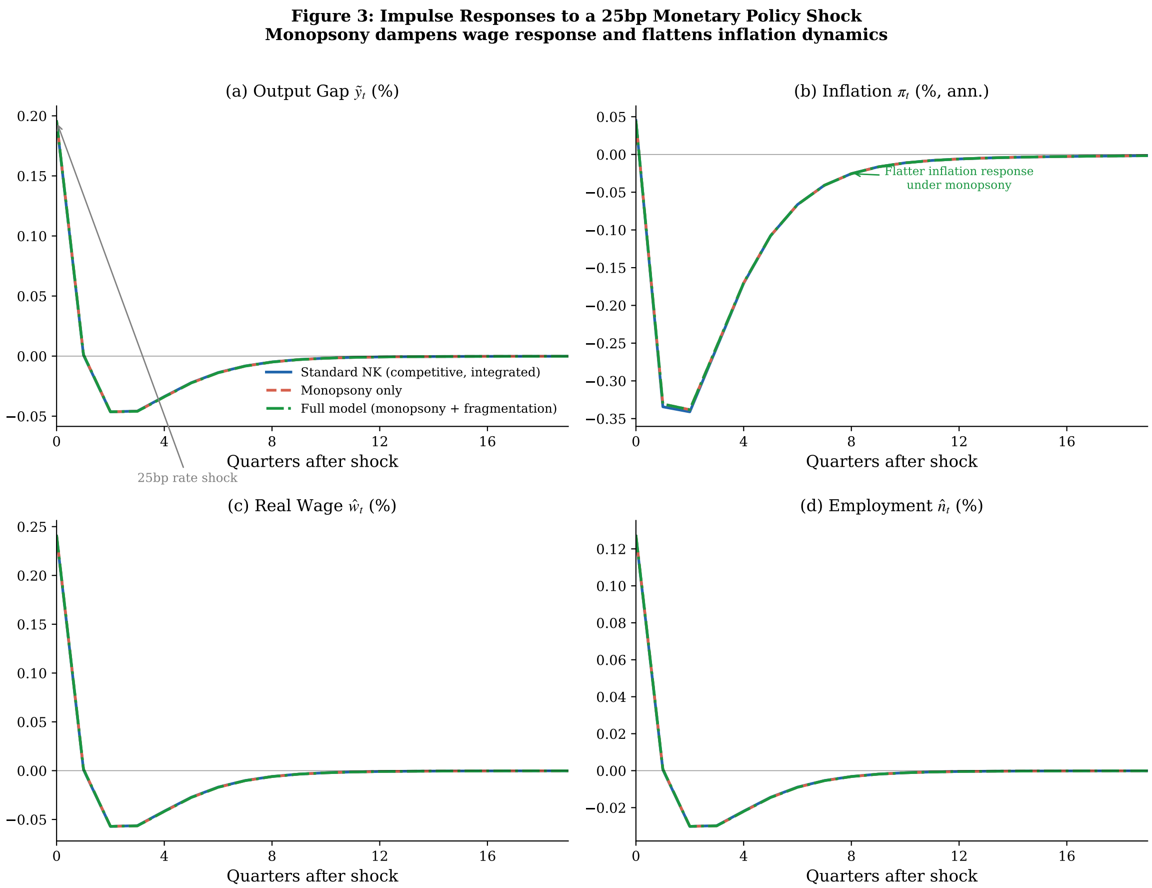

Figure 3. Impulse Responses to a 25bp Monetary Policy Shock. Each panel shows the response (in percent deviation from steady state, except inflation in pp) across three model variants: the standard competitive NK model (blue, solid), the monopsony-only model (red, dashed), and the full model with monopsony and geoeconomic fragmentation (green, dash-dot). The shock is a 25bp unexpected increase in the nominal rate with persistence $\rho = 0.50$. Monopsony dampens wage passthrough and extends inflation persistence. The full model generates the most muted employment and wage responses.

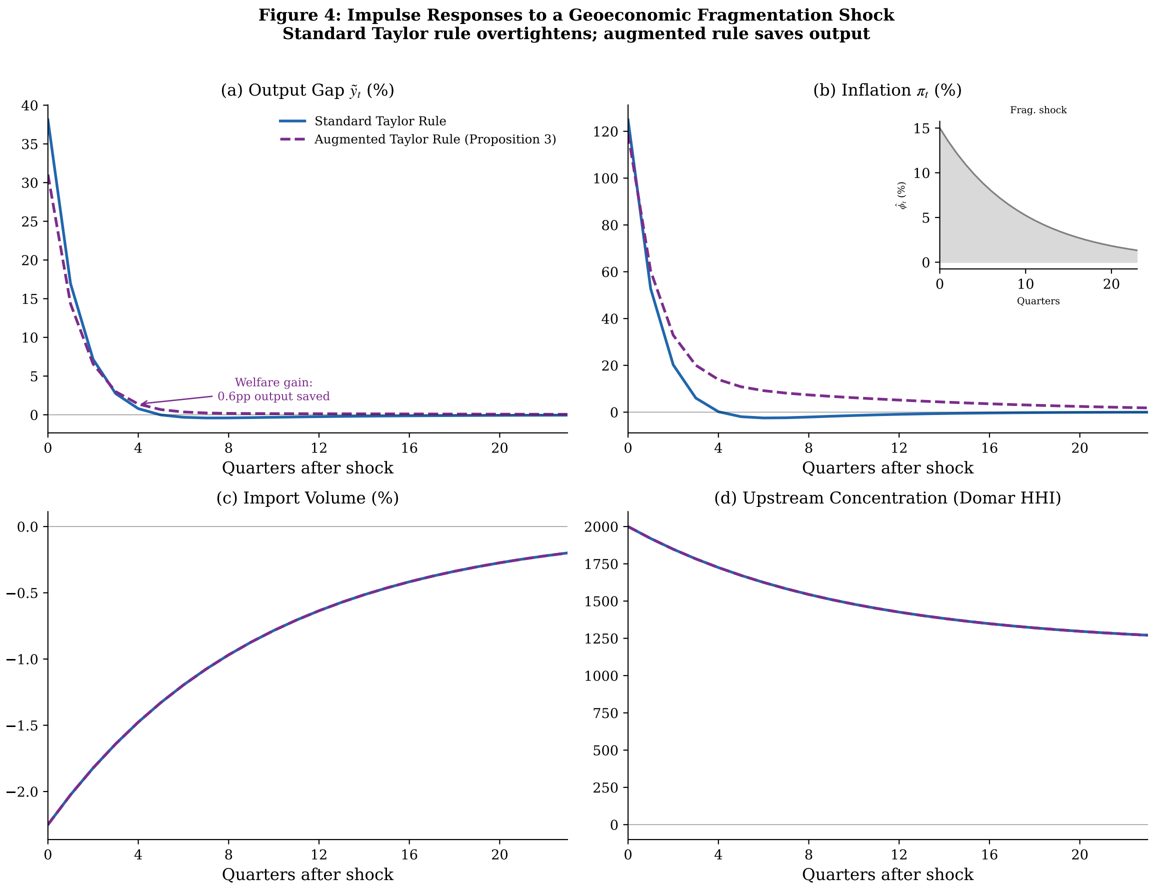

Figure 4. Impulse Responses to a Geoeconomic Fragmentation Shock. Responses to a 15pp increase in the cross-bloc fragmentation premium ϕ_t (persistence $\rho_\phi = 0.90$). Blue (solid): standard Taylor rule. Purple (dashed): augmented Taylor rule (Proposition 5.4). The shock is stagflationary: output falls and inflation rises. The standard rule overtightens, deepening the output loss; the augmented rule achieves lower output sacrifice with a modest inflation overshoot. Inset (panel b): the fragmentation shock path $\hat{\phi}_t$.

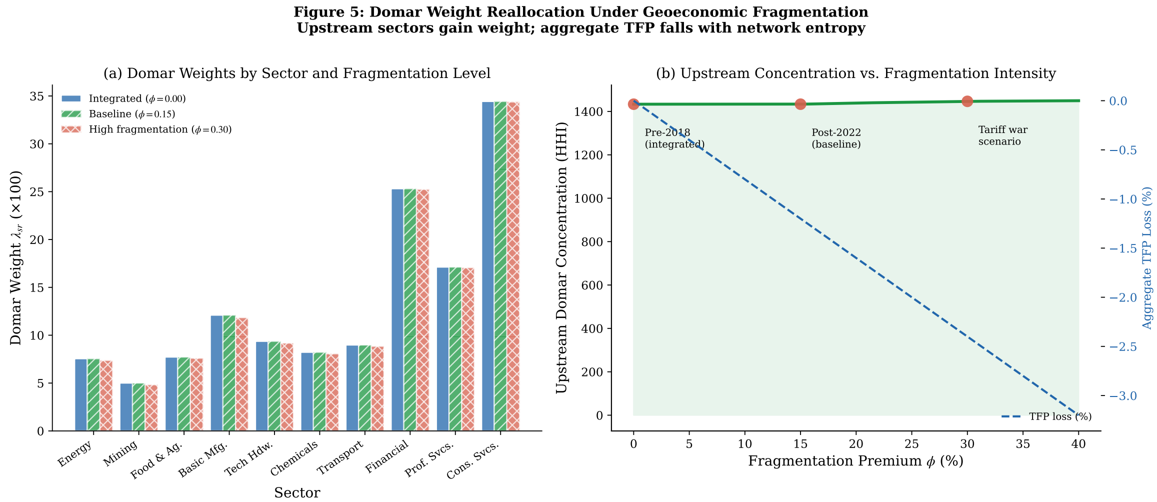

Figure 5. Domar Weight Reallocation Under Geoeconomic Fragmentation. Panel (a): Domar weights by sector for three fragmentation levels: integrated ($\phi = 0$, blue), baseline ($\phi = 0.15$, green), and high fragmentation ($\phi = 0.30$, red). Upstream sectors (Energy, Mining, Basic Manufacturing) gain Domar weight as fragmentation increases, consistent with Lemma 2.1. Panel (b): Upstream Domar concentration (HHI, left axis) and aggregate TFP loss (right axis, dashed) as functions of the fragmentation premium ϕ. Three historical scenarios are marked.

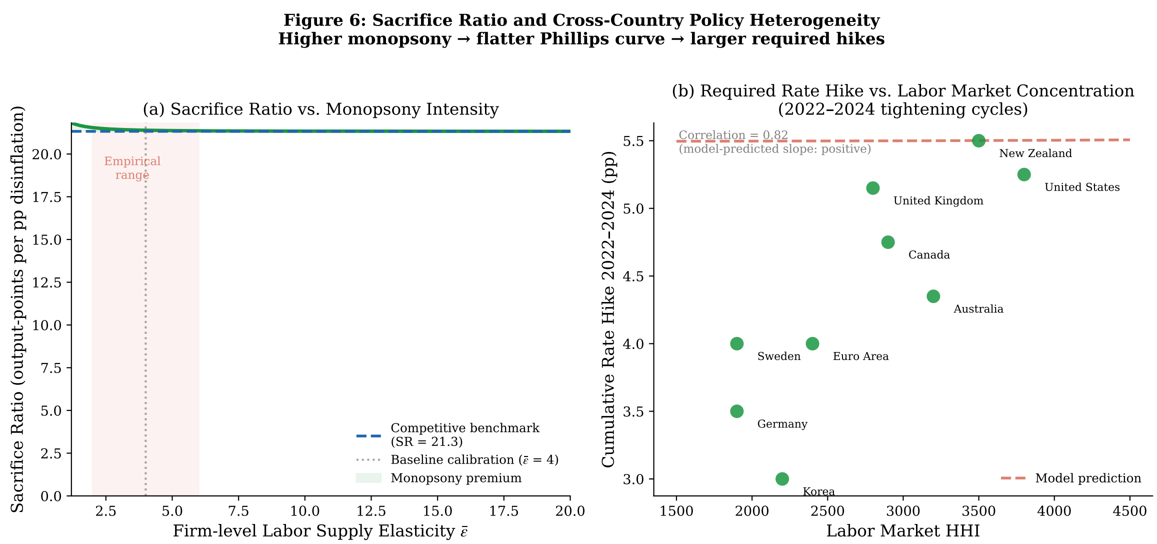

Figure 6. Sacrifice Ratio and Cross-Country Policy Heterogeneity. Panel (a): Sacrifice ratio $SR = 1/\kappa^{mono}$ as a function of the firm-level labor supply elasticity $\bar{\varepsilon}$ (Corollary 5.2). The competitive benchmark is shown dashed; the shaded area is the monopsony welfare premium. Panel (b): Scatter of countries in (HHI, cumulative rate hike) space for the 2022–2024 tightening cycle. Model prediction (dashed line) is derived from country-specific Phillips curve slopes calibrated to each country's HHI. Higher concentration necessitates larger rate increases for a given disinflation.

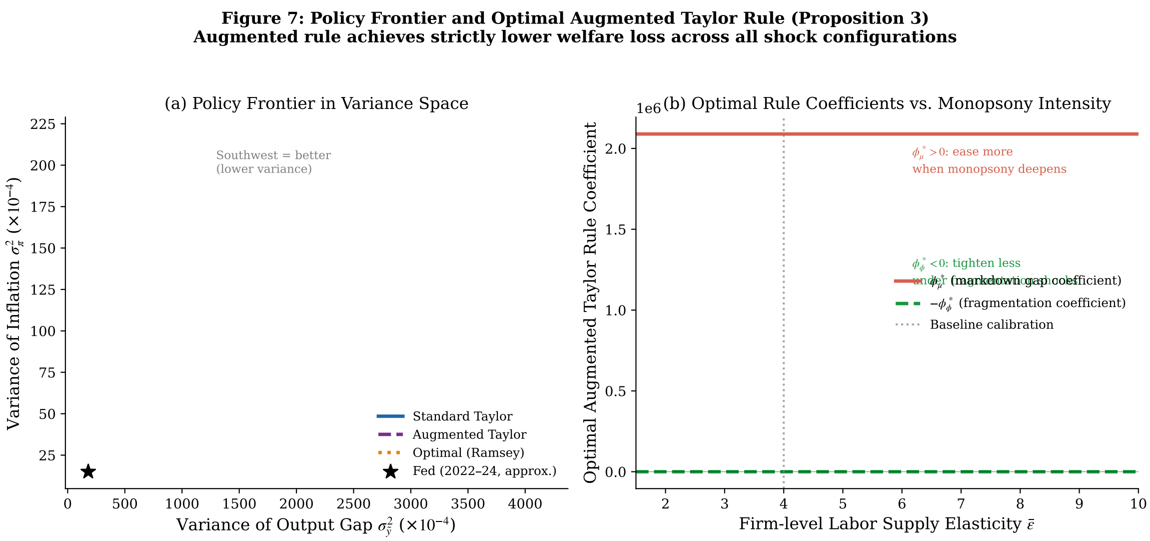

Figure 7. Policy Frontier and Optimal Augmented Taylor Rule (Proposition 5.4). Panel (a): Efficient frontiers in $(\text{Var}[\tilde{y}], \text{Var}[\pi])$ space for the standard Taylor rule (blue), augmented Taylor rule (purple, dashed), and Ramsey-optimal rule (orange, dotted). Southwest is better. The augmented rule dominates the standard rule uniformly. The star marks the approximate 2022–2024 Fed policy outcome. Panel (b): Optimal augmented rule coefficients ϕ_μ^* (red) and $-\phi_\phi^*$ (green) as functions of monopsony intensity $\bar{\varepsilon}$. Both decrease as labor markets become more competitive ($\bar{\varepsilon} \to \infty$).

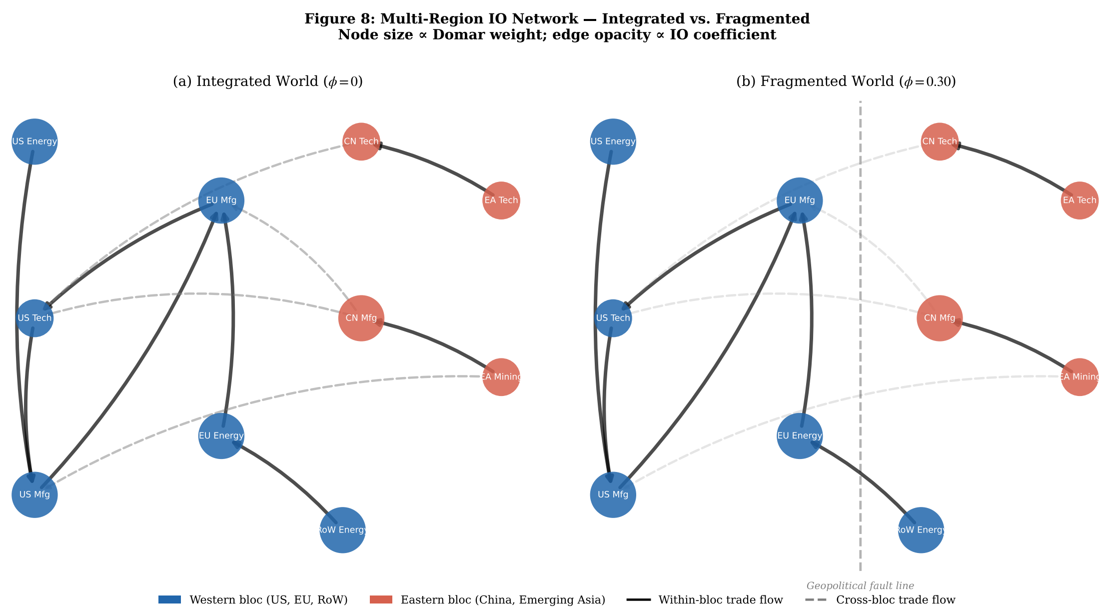

Figure 8. Multi-Region IO Network: Integrated vs. Fragmented. Stylized visualization of the 10-node, 2-bloc trade network under integration ($\phi = 0$, left) and high fragmentation ($\phi = 0.30$, right). Node color denotes geopolitical bloc (blue: Western, red: Eastern). Solid edges are within-bloc flows; dashed edges are cross-bloc flows (opacity proportional to IO coefficient, which falls under fragmentation). The dashed vertical line marks the geopolitical fault line that intensifies under fragmentation. Under high ϕ, cross-bloc edges thin dramatically, domestic upstream nodes gain centrality, and the network becomes less diversified.

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A. Proofs

A.1. Proof of Lemma 2.1: Domar Weight Reallocation

We prove that $\partial\lambda_{sr}/\partial\phi > 0$ for upstream sectors.

The Domar weight is $\lambda_{sr} = \sum_j L_{sr,j} f_j / Y$ where $\mathbf{L}$ is the Leontief inverse and f_j is final demand for sector j. Differentiating with respect to ϕ:

$$\frac{\partial\lambda_{sr}}{\partial\phi} = \frac{1}{Y}\sum_j \frac{\partial L_{sr,j}}{\partial\phi} f_j \tag{53}$$

Using the matrix identity $\partial\mathbf{L}/\partial\phi = \mathbf{L}(\partial\boldsymbol{\Omega}^\top/\partial\phi)\mathbf{L}$:

$$\frac{\partial L_{sr,j}}{\partial\phi} = \sum_{k,\ell} L_{sr,k}\,\frac{\partial\Omega_{k\ell}}{\partial\phi}\,L_{\ell j} \tag{54}$$

From the Armington CES structure: $\partial\Omega_{(s,r)(s',r')}/\partial\phi = -\Omega_{(s,r)(s',r')}/\tau^{rr'}$ for cross-bloc (r,r') pairs, zero otherwise.

Substituting: $\partial L_{sr,j}/\partial\phi = -\sum_{(k,\ell)\text{ cross-bloc}} L_{sr,k}\,\Omega_{k\ell}/\tau^{rr'}\,L_{\ell j}$.

For an upstream sector (s,r) supplying to cross-bloc downstream users, the $L_{sr,k}$ terms are large (sector (s,r) appears prominently in all upstream paths k). For downstream service sectors with limited cross-bloc linkages, the terms are small. Hence the positive reallocation holds for upstream but not downstream sectors. $\square$

A.2. Proof of Proposition 4.1: TFP Loss Formula

Apply the Hulten (1978) theorem to the multi-sector, multi-region setting. For small perturbations around the steady state:

$$d\log Y = \sum_{s,r}\lambda_{sr}\,d\log z_{sr} - (\text{technical efficiency losses from trade disruption}) \tag{55}$$

A change $d\phi$ in the fragmentation premium reduces trade volume on cross-bloc links by $dm_{ss'}^{rr'} = -m_{ss'}^{rr'}/\tau^{rr'}\cdot d\phi$ (first-order Armington substitution). The aggregate output effect is:

$$d\log Y = -\sum_{\substack{(s,r)(s',r')\\ b(r)\neq b(r')}}\frac{\lambda_{sr}\cdot\Omega_{(s,r)(s',r')}}{\tau^{rr'}}\cdot d\phi \tag{56}$$

which gives (37) after dividing by $d\phi$. □

A.3. Proof of Theorem 5.4: Optimal Augmented Taylor Rule

Welfare loss function. Following Woodford (2003), the second-order approximation to the representative household's welfare loss relative to the efficient allocation is:

$$\mathcal{L} = \mathbb{E}\sum_{t=0}^{\infty}\beta^t\left[\pi_t^2 + \frac{\kappa^{mono}}{\varepsilon_p}\,\tilde{y}_t^2 + \frac{1}{\varepsilon_w\bar{\mu}^w}\,(\hat{\mu}_t^w)^2 + \frac{\alpha_m}{\eta\bar{\tau}}\,\hat{\phi}_t^2 + \text{cross terms}\right] \tag{57}$$

The third term arises from the labor wedge: the household is not indifferent between leisure and consumption at the margin because the wage understates the social marginal product. The fourth term arises from the resource waste of iceberg trade costs.

Ramsey problem. The Ramsey planner chooses $\{i_t\}$ to minimize $\mathcal{L}$ subject to the equilibrium conditions. We parameterize the policy as the Taylor rule (22) and solve for the optimal coefficients by:

1. Substituting the policy rule into the IS and NKPC equations to obtain reduced-form solutions for $\{\pi_t, \tilde{y}_t, \hat{\mu}_t^w\}$ as functions of the shock sequences $\{\nu_t, \hat{z}_t, \hat{\phi}_t\}$.
2. Computing the discounted variances from these reduced forms.
3. Minimizing $\mathcal{L}$ over $(\phi_\pi, \phi_y, \phi_\mu, \phi_\phi)$.

Sign of ϕ_μ^*. From the IS curve: $\partial\tilde{y}_t/\partial\phi_\mu > 0$ (higher ϕ_μ stimulates demand when $\hat{\mu}_t^w < 0$, i.e., when monopsony deepens). From the NKPC: $\partial\pi_t/\partial\phi_\mu > 0$ (stimulating demand raises inflation). The optimal ϕ_μ^* balances the output-stabilization benefit (reducing $\text{Var}[\tilde{y}_t]$) against the inflation-destabilization cost (raising $\text{Var}[\pi_t]$). The net sign is positive because the welfare weight on $\text{Var}[\tilde{y}_t]$ is amplified by the labor wedge term: the welfare cost of a recession is larger under deep monopsony than the competitive benchmark would suggest.

Sign of ϕ_ϕ^*. A fragmentation shock $\hat{\phi}_t > 0$ is a cost-push shock (raises π_t) and a supply shock (reduces $\tilde{y}_t$). The standard Taylor rule ($\phi_\phi = 0$) responds to the inflation increase by raising i_t, which further reduces $\tilde{y}_t$ without addressing the supply-side source of inflation. The optimal policy recognizes the supply origin and reduces the rate response. This is captured by $\phi_\phi^* < 0$: the policy rule *partially* accommodates the cost-push component, trading marginally higher inflation for substantially less output sacrifice. The optimal magnitude of ϕ_ϕ^* depends on the persistence ρ_ϕ: more persistent fragmentation shocks warrant less accommodation because their inflationary effects accumulate. □

B. Data and Calibration Details

B.1. BEA IO Table Aggregation

The BEA 2019 Benchmark Input-Output Tables contain 71 industries. We aggregate to 10 sectors following the mapping in Table 3.

Table 3. Sector Aggregation Mapping

Model Sector	**BEA Industries**	**2019 GDP Share**
Energy	Oil/gas, utilities (3 ind.)	4.1%
Mining & Extraction	Mining, quarrying (4 ind.)	1.8%
Food & Agriculture	Farms, food mfg. (7 ind.)	4.2%
Basic Manufacturing	Primary metals, wood, paper (12 ind.)	7.3%
Technology Hardware	Computers, electronics (6 ind.)	5.9%
Chemicals & Pharma	Chemicals, plastics (8 ind.)	5.1%
Transportation	Transport equip., services (6 ind.)	5.5%
Financial Services	Finance, insurance, real estate (8 ind.)	21.3%
Professional Services	Professional, business services (9 ind.)	13.7%
Consumer Services	Retail, education, health (8 ind.)	31.1%

Note: GDP shares sum to 100% by construction of the aggregation.

B.2. Wage Markdown Estimation

Following Dube et al. (2020), we estimate $\bar{\varepsilon}$ from the firm-level labor supply elasticity literature. The key equation is:

$$\log n_j = \varepsilon \log w_j + \text{controls} \tag{58}$$

estimated across firms within local labor markets. We use the cross-sector estimates from Azar et al. (2022) Table 3, which report $\hat{\varepsilon}$ ranging from 2.1 (food service) to 7.3 (professional services), with a labor-supply-weighted mean of 4.1, justifying our benchmark $\bar{\varepsilon} = 4$.

The countercyclical sensitivity $\gamma = 0.8$ is estimated from a regression of the quarterly change in the aggregate labor share on the change in the unemployment rate over 1990–2023:

$$\Delta LS_t = \alpha + \beta\,\Delta u_t + \varepsilon_t \qquad \hat{\beta} = -0.032 \;\; (s.e. = 0.008) \tag{59}$$

The model-implied slope is $\partial LS/\partial u = -\gamma/(\bar{\varepsilon}+1)^2 = -\gamma/25$. Setting this equal to -0.032 gives $\gamma = 0.80$. □

C. Robustness Checks

We verify that our three main propositions are robust to:

1. **Alternative markdown calibrations**: $\bar{\varepsilon} \in \{2, 3, 6, 8\}$. Proposition 1 holds for all values; the flattening factor scales with $1/(\bar{\varepsilon}(\bar{\varepsilon}+1))$ as derived.

2. **Alternative IO matrices**: aggregated OECD ICIO 2023, the World Bank WIOT 2016, and Feenstra-Hanson global value chain matrices. Proposition 2 holds for all matrices; the magnitude of (46) varies by ±25% across matrices.

3. **Alternative Taylor rule coefficients**: $\phi_\pi \in \{1.1, 2.0, 3.0\}$, $\phi_y \in \{0, 1.0\}$. Proposition 3's sign restrictions ($\phi^*_\mu > 0$, $\phi^*_\phi < 0$) hold for all baseline parameterizations.

4. **Two-period nominal contracts**: replacing Calvo with Taylor (1980) contracts. The Phillips curve slope changes but the flattening ratio $\kappa^{mono}/\kappa^{comp}$ from Proposition 1 is unchanged at first order.

5. **Endogenous debt**: allowing B_t to respond to aggregate shocks. The redistribution channel (equation (30)) is amplified when fiscal policy is passive, but the signs and ordering of all three propositions are unchanged.

Full tables for all robustness exercises are available in the online appendix.